Berner Fachhochschule
Haute école spécialisée bernoise
Bern University of Applied Sciences

Package for Laurin

Memo, slides and peer evaluation: everything in one PDF

Item	Details
Institution	Bern University of Applied Sciences (BFH)
Course	EAIF: AI for Finance, Autumn Semester 2026
Group	Group_8 (Beyeler Fabian, Lussi Laurin, Griessen Jan)
Document type	Combined package of 8 documents in one PDF
Key dates	Coaching Tue 24 Nov 2026; memo, notebook, results.json and peer evaluation Fri 27 Nov 2026, 23:59; slides Mon 7 Dec 2026, 23:59; presentation Tue 8 Dec 2026
Status	4 November 2026

> **Purpose.** Everything Laurin needs to write the memo and the slides, in the order in which it is most useful.

Contents

Document	Pages
1. Detailed outline for Laurin (memo, slides, Q&A)	2 to 11
2. Team briefing (rules and deadlines)	12 to 14
3. Task (a) results sheet: audit and pipeline	15 to 18
4. Task (b) results sheet: models	19 to 22
5. Task (c) results sheet: where ESG pays	23 to 26
6. Task (d) results sheet: backtest	27 to 30
7. Plain-language explainer (background)	31 to 37
8. Group plan and timeline	38 to 41

Suggested reading order

- **First:** the outline (part 1) and the briefing (part 2): what the memo must contain and which rules decide points.
- **Then:** the four results sheets (parts 3 to 6): numbers, figures, suggested text and likely questions for each task.
- **When needed:** the explainer (part 7) for everyday-language background, and the plan (part 8) for timeline and roles.
- **Questions:** ask Fabian before writing down any number that looks odd. The notebook and the figures for the slides are in the repository folder group_project_AS2026.

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Detailed outline for Laurin

Memo, slides, questions and peer evaluation

Item	Details
Institution	Bern University of Applied Sciences (BFH)
Course	EAIF: AI for Finance, Autumn Semester 2026
Group	Group_8 (Beyeler Fabian, Lussi Laurin, Griessen Jan)
Document type	Working outline for the memo (3 pages plus appendix), the slides (8 minutes) and the peer evaluation
Key dates	Coaching Tue 24 Nov 2026; memo, notebook, results.json and peer evaluation Fri 27 Nov 2026, 23:59; slides Mon 7 Dec 2026, 23:59; presentation Tue 8 Dec 2026
Source of numbers	Executed solution notebook, local run of 4 November 2026 (a fresh Colab run follows before submission)

Purpose. Tells you what each page of the memo must contain, which numbers to use and where each is computed, wording you can edit, and what not to say. The analysis is finished.

How to use this outline. It tells you what each page of the memo must contain, which numbers to use (and where each one is computed in the notebook), wording you can edit, and what not to say. The four results sheets (tasks a to d, in the folder results_sheets/) hold the detailed numbers, figures and explanations; the plan is in docs/PLAN.md. You do not need to code. If a number looks odd, ask Fabian before you write it down.

1. Deadlines and hand-in checklist

Date	What	Notes
Sun 8 Nov	Memo skeleton in Word (the six headings) and slide skeleton (6 slides)	Walkthrough with Fabian, 30 minutes
Sun 15 Nov	Memo sections 1 to 4 drafted; outline of section 5	Use this outline
Sun 22 Nov	**Complete memo draft and complete deck**	Notebook is already frozen in content
Mon 23 Nov	Team read-through, list of questions for the coaching	Everyone reads everything
Tue 24 Nov	**Coaching with the lecturer**	Bring draft memo and deck
Thu 26 Nov	Final memo PDF; peer evaluation filled in by each of us; **submit in the evening**	Hard deadline Fri 27 Nov, 23:59
Mon 7 Dec	Final slides (PDF), internal target noon, deadline 23:59	Moodle or email to the lecturers
Sat 5 and Mon 7 Dec	Two timed rehearsals: 8 minutes	Questions afterwards: about 2 minutes
Tue 8 Dec	Presentation	Everyone answers questions on every slide

- **Four deliverables carry the group name "Group_8":** the memo PDF (name and members on page 1), the notebook, the slides PDF and results.json (the notebook writes the group name into it). Suggested file names: Group_8_memo.pdf, Group_8_notebook.ipynb, Group_8_slides.pdf, Group_8_results.json.
- **Moodle:** memo, notebook, results.json and the peer evaluation (one form per student) by Fri 27 Nov, 23:59. Slides by Mon 7 Dec, 23:59. Questions to the lecturers go to the Moodle forum.

- **Fabian** produces the final notebook run and results.json; **you** produce the memo and the slides; each of us fills in the peer evaluation.

2. What earns points (rubric) and where it goes in the memo

Criterion (points)	What the grader looks for	Where in the memo
Pipeline and data audit (20)	5: leakage-free formation panel. 15: the audit table, graded against the list of planted problems	Section 3 (compact table) and appendix A1, A2
Model comparison (20)	Linear vs tree ensemble, selection on cross-validation, test reported once, next to a naive baseline	Section 4(b), appendix A4
Where does ESG pay (15)	8: finding the cell where the premium exists and only that cell. 4: honest uncertainty. 3: point-in-time vs restated	Section 4(c), appendix A5
Backtest with costs and floor (15)	Formed after publication, held 12 months, costs charged, cost of the floor shown, bootstrap of the difference to the benchmark	Section 4(d), appendix A6
Memo and slides (15)	Clear question, clear answer, figures with labelled axes and units, no overclaiming; 10-minute talk (we plan 8)	Whole memo, the slides
What would change with real data (15)	5: the list of what remains after the audit. 10: which conclusions would survive and which not, and why	Section 5 (about 0.6 page)

Three rules the handout states: every metric appears next to the baseline it must beat (the naive average forecast for models, the equal-weight benchmark for the backtest); every figure has labelled axes with units; every feature is dated. **Also:** state that the data is simulated and cite it with its licence on the memo and on the slides: "Simulated ESG and stock-return dataset, EAIF course, Autumn Semester 2026, CC BY 4.0", https://github.com/umatter/EDFB/tree/main/data/esg2026.

3. Memo: page budget (3 pages including tables and figures, plus an appendix of any length)

The lecturer confirmed that tables and figures count towards the three pages. Use 10 pt text, 2 cm margins, 8 pt tables, and put everything else in the appendix with a pointer ("see Appendix A4").

Page	Content	Space
1	Title block (Group_8, members, date), 1 Question and short answer, 2 Data, start of 3 Audit: compact audit table (7 rows)	Title 4 lines; Question 5 lines; Data 8 lines; table 0.4 page
2	3 Pipeline: decision rule, split, feature table (compact), 4(b) model comparison (text + 4-row table), start of 4(c)	Pipeline 0.2 page; feature table 0.25; (b) 0.3; (c) start 0.2
3	4(c) region table (compact), 4(d) text + 4-row table, 5 What would change with real data, 6 Group members and roles	(c) 0.2; (d) 0.25; real data 0.45; roles 3 lines
Appendix	A1 full audit table; A2 full feature table; A3 figures a1, b1, b2; A4 tuning grid and full (b) table; A5 figures c1, c2 and full (c) table; A6 figure d1 and the robustness tables; A7 how to reproduce (Colab link, run order)	No limit

If page 3 is too full, shorten section 5 by moving the "list" into the table of conclusions, but keep both halves: 5 points are for the list, 10 for which conclusions survive and why.

Word budget. With the tables and one figure, three pages hold about **1,200 words of running text.** The suggested texts in this outline and on the results sheets add up to more than that on purpose: they give you material to choose from, not text to paste in full. Keep to the budget below and move the rest to the appendix or to your notes for the questions.

Section	Words	Tables
Title block, 1 Question and short answer	120	
2 Data	80	
3 Audit (two sentences around the table) and pipeline	180	Audit table (4 columns, 7 rows); compact feature table
4(b) Model comparison	120	Train / CV / test table (4 rows)
4(c) Where ESG pays	130	Region by period table (8 rows)
4(d) Backtest with costs and the floor	140	Backtest table (4 rows)
5 What would change with real data	380	Optional: a compact survive / not table
6 Group members and roles	40	
Total	**about 1,190**	

4. Memo, section by section

Title block (page 1, top)

- "Group_8: Can ESG scores and machine learning earn an investor a premium? A memo to the investment committee". Members: Beyeler Fabian, Lussi Laurin, Griessen Jan. Date. One line: "All data are simulated.".

1. The question (5 lines, plus a two-line answer)

- **Question:** "Can an investor use ESG scores, or forecasts built on them, to earn higher returns, and can the data be trusted for that purpose?" Four sub-questions: is the data reliable, can a model forecast 12-month returns better than the average, where (if anywhere) did the Environmental score pay, and does a rule with costs and an ESG floor beat an equal-weight benchmark?
- **Short answer (committee readers love this):** "The data needed seven fixes, one of which (a leaking column) would have produced a false forecasting edge. No model beat the naive forecast. The Environmental score was associated with higher returns in one place only, Europe from 2021, on four years of data. A backtest with costs showed no edge that can be told apart from noise. We would not allocate on this evidence."

2. The data (8 lines)

- Simulated by the course, no real firms. Five files: firm-year panel (14,250 rows, fundamentals, point-in-time ESG scores, environmental metrics), restated ESG scores (as a current download would show them), monthly returns (171,764 firm-months), monthly factors, and the current universe (987 survivors). 1,376 firms, fiscal years 2012 to 2025; firms enter and leave (389 exits).
- Licence line, exactly: "Simulated ESG and stock-return dataset, EAIF course, Autumn Semester 2026, CC BY 4.0", with the link.
- Say in one sentence that simulated data cannot show how ESG behaves in real markets; it tests our method. (This returns in section 5.)

3. The audit and the pipeline (page 1 bottom to page 2 top)

Audit table: the four columns the handout requires (issue, how detected, how treated, effect on one headline number). Use short phrases; the long version is appendix A1. Rows and numbers (source: notebook (a) Step 10):

Issue	How detected	How treated	Effect on one headline number
28 firms: revenue and market cap about 1,000x too large (probably USD thousand)	Carbon, water and energy out of line with revenue for the same firms	Divide by 1,000	Mean market cap 213,205 to 14,108 USD million

Issue	How detected	How treated	Effect on one headline number
300 near-duplicate return months	Duplicate company-month keys	Keep first row	Firm-years with a full 12 months: 9,002 to 9,182
Analyst_Outlook carries future information	Rank correlation with the return in three 12-month windows: 0.68 with the next, about 0 with the others	Exclude from all models	Random forest test error 37.6 without it, 29.1 with it (baseline 37.2)
Survivorship: universe file has 987 of 1,376 firms	File sizes; 389 firms end before 2025	Use the full panel, hold exits to their last month	Equal-weight benchmark 4.62% (full panel) vs 5.18% (survivors only), formation years 2015 to 2025
Restated scores behave as if they contain hindsight	Comparison of the two score files with later returns	Point-in-time scores everywhere; restated only for the comparison	Within-year rank correlation with next excess return 0.010 vs 0.056
Carbon, water, energy missing together (13.2% of rows), not at random	Missing share by region and by score	Keep rows, carbon intensity filled plus a missing flag	Linear test error 38.10 to 38.05 without carbon features
Scores clipped at 0 or 100 (213 rows, 1.5%), 8 returns at -95%	Counts at the bounds	Keep, document	Linear test error 38.10 to 38.00 without clipped rows

- Add the sentence from the notebook below the table: "Each effect was measured one problem at a time, so the effects do not add up."
- **Pipeline (5 to 6 lines).** Portfolios are formed on 1 July of each year; every feature comes from the fiscal-year row published between 31 March and 31 May (known before 1 July) or from returns up to June; the target is the total return over the next 12 months. Split by formation year: training 2015 to 2021, test 2022 to 2024 (the test set is used once); cross-validation by company inside the training years. Mention the decision-time check built into the code: nothing is used before its publication date.
- **Feature table (compact, 8 pt):** group the given features in one row ("given: growth, margin, four ESG scores, size, past 12-month return, industry and region"), then one row each for the six new features with the date from which each is known: Env_Change, Margin_Change, log_MC_to_Rev, log_Carbon_Int, Carbon_Missing (all from the fiscal-year row, known by 31 May of the formation year) and Past_Vol_Pct (returns through 30 June). Analyst_Outlook as the row "excluded". Full table: appendix A2 (notebook (a) Step 9).

4(b). Model comparison (about 0.35 page)

- **Text:** use the memo paragraph on results sheet (b). The handout asks for train, CV and test error next to the baseline, so the table needs all three columns:

Model	Train RMSE	CV RMSE	Test RMSE
Naive mean (baseline)	29.16	29.16	37.20
Linear regression (selected on CV)	28.73	28.88	38.10
Random forest (tuned)	28.11	28.92	37.93
Gradient boosting (tuned)	28.28	28.90	37.84

- **The one sentence the handout wants:** "The extra complexity of the ensembles does not pay off out of sample."
- **Must-have honesty notes:** (1) the selection of linear regression is effectively a tie (within 0.05 points; the best model changes from fold to fold), and linear regression has nothing to tune; (2) the test years were a +21% boom (against +0.9% in training), so cross-validation (28.9) looks far better than the test (37 to 38); the starter notebook's "downturn" is wrong for this data; (3) after removing year averages the models still do not beat zero, rank correlation about -0.02 and changing sign; (4) validated in time order inside the training years, no model beats the naive forecast either ((b) Step 6), which fits the test result.
- **Translate for the committee:** an error of 37 to 38 points on 12-month returns means these forecasts cannot size or rank positions.

4(c). Where does ESG pay (about 0.4 page)

- **Text:** use the memo paragraph on results sheet (c). The deliverable is one table (region by period, point-in-time and restated) and one sentence. Compact table: 7 regions plus "All regions", four columns, coefficient with standard error in brackets, stars. Full table with intervals and p-values: appendix A5.
- **The sentence (copy from results.json):** "An investor could have earned an ESG premium only in firms in Europe, from 2021 (+3.1 points of 12-month excess return per standard deviation of the Environmental score; Holm-adjusted p = 0.018 with errors clustered by company, but p = 0.09 with its 4 formation years as the unit); no other region or period is distinguishable from zero, so this is a plausible, not a proven, premium, and it is a cross-sectional association, not a return shown to be tradable after costs."
- **Honest uncertainty (4 points):** 1 of 14 cells; Holm 0.018 holds only with errors clustered by company, with the four formation years as the unit p is about 0.09; intervals not adjusted for the 14 cells; the premium shows from 2022 (2021 is +0.35).
- **Point-in-time vs restated (3 points):** restated scores give higher coefficients everywhere (by 0.9 to 2.5 points) and, before the Holm correction, more significant cells (3 against 1). After the correction both versions keep only Europe from 2021, at +4.88 restated against +3.08 point-in-time. The reason: the restatement behaves as if it contains hindsight (what it added has a rank correlation of 0.16 with the next return). Say "behaves as if", not "contains".

4(d). Backtest with costs and an ESG floor (about 0.3 page)

- **Text:** use the memo paragraph on results sheet (d). Compact table with the four rows (no floor, floor 50th, floor 80th, benchmark): return, volatility, Sharpe, turnover. Figure d1 goes to appendix A6; refer to it.
- **The sentence:** "The floor did not cost return in this sample (it cost concentration and trading), but the gain is not an ESG effect: every firm above the 80th percentile at equal weight earns about the benchmark (8.9% against 8.8%), and without formation year 2022 every rule trails the benchmark. The best rule's Sharpe advantage (+0.11, 95% interval -0.16 to +0.38) is not distinguishable from noise."
- **Must-have notes:** the model used is the one selected in (b) and has almost no ranking skill (rank correlation +0.01); 11% to 19% of random selections of the same size do as well as the rules; the 80th-percentile floor holds about 19 names; formation year 2025 (six months) is left out, and including it lowers every rule by about 2 points without changing the ranking; the clean out-of-sample years 2022 to 2024 tell the same story; gradient boosting and a linear variant are labelled robustness checks; costs are the starter's 20 bp convention (40 bp on the replaced share); benchmark is equal weight, no cost, full panel including exits.

5. What would change with real data (about 0.6 page; 15 points). See the detailed content in the next section.

6. Group members and their roles (3 lines)

- Beyeler Fabian: notebook (data lead and modelling lead). Lussi Laurin: writing lead (memo and slides). Griessen Jan: **role to be agreed by the group before the memo is finished** (the handout asks for roles to be named; a natural fit is checking tables and figures against the notebook). Close with the attribution line again.

5. Detailed content for "What would change with real data"

5a. The list: what remains to be solved after an audit like ours (5 points)

Write these as a compact list, one line each; the rightmost column is evidence from our own work you can quote.

What remains unsolved on real data	Why it matters	Evidence in our work
ESG scores are proprietary and raters disagree	Our data has one score per firm. Real providers use different methods and their scores correlate only moderately (published research reports an average correlation around 0.5 between major raters: Berg, Koelbel and Rigobon 2022; verify the figure before citing). A result for one rater may not hold for another.	Restated vs point-in-time: 86% overlap in the top decile, so 14% of the names change
Point-in-time availability	Vendors revise and backfill scores. Without dated vintages we cannot know what an investor saw on 1 July. We had a publication date and a separate restated file; real data rarely does, and an Analyst_Outlook-type column would be hard to spot.	Restated scores behave as if they contain hindsight; Analyst_Outlook: test error 37.6 to 29.1
Survivorship that is harder to see	Our universe file betrayed itself (987 of 1,376 firms). Real databases silently drop delisted firms, and delisting returns (bankruptcy, takeover) are often missing.	Survivors-only benchmark overstates the return by about 0.6 points a year (2015 to 2025; 0.46 over the 2018 to 2024 backtest); -95% returns not in the last month
Costs are not a constant	We used 20 bp each way. Real costs depend on spread, size, liquidity, market stress and the size of the order; a concentrated 19-name portfolio is harder to trade.	Turnover 0.70 to 0.82; cost drag 0.28 to 0.33 points a year
Units, identifiers and duplicates	We found a unit error and duplicates through cross-column checks. Real data has mergers, name changes, currency and restated fundamentals that are harder to detect.	28 firms off by a factor of 1,000; 300 duplicated months
Missing ESG data is not random	Companies choose what to disclose. Missingness related to the score and to region biases any sample that drops gaps, and it is lowest in Europe, the one region where we found an effect.	13.2% missing; Europe 6.3% vs 13 to 19% elsewhere
Short samples, one regime, many looks	14 years, a three-year test boom, 14 region-period cells. Real data is longer but the risk of a false discovery remains.	Test years +21.6% vs +0.9%; Europe result rests on four years (p about 0.09)

5b. Which of our conclusions would survive, which not, and why (10 points)

This is the larger part. Use a table: conclusion, survives or not, why, and what would test it on real data.

Our conclusion	On real data	Why	What would test it
No model beats the naive forecast out of sample	**Likely survives in direction**	Stock returns are weakly predictable; the signal-to-noise ratio is low everywhere. With real data and more years a small skill could appear, so the claim is "no evidence of skill", not "impossible".	Time-forward cross-validation over many years; several regimes; more features
The specific data problems (unit error, duplicates, a leaking column)	**Do not survive as such**	They were planted. Real data has other problems. What survives is the procedure: consistency checks across columns, correlation with returns in several time windows, publication dates, comparison with a survivors-only file.	Run the same checks on the real vendor files and ask the vendor for documentation
Survivors-only and restated data flatter results	**Direction survives, size does not**	Both mechanisms are real; their size depends on the exit rate, delisting returns and vendor practice. The 0.6 points a year is specific to this simulation.	Point-in-time constituent lists, delisting returns, data vintages
Environmental score paid in Europe from 2021 (+3.1 points per SD)	**Does not survive as an investable conclusion**	We cannot tell whether the cell reflects a relationship built into the simulation or chance (p about 0.09 with years as the unit). In real data a single cell out of 14, on four years, would need replication before anyone relied on it. The result may also reflect disclosure rules or regional events we cannot see here.	Out-of-sample replication on other years, other raters and other regions; a pre-registered hypothesis

Our conclusion	On real data	Why	What would test it
Restated scores overstate the premium	**Likely survives in direction**	Backfilled or revised scores can embed information that arrived later; any test on them is optimistic. The size cannot be known without vintages.	Compare strategies built on original versus revised scores
Backtest: no edge distinguishable from the benchmark	**Survives, and net results get worse**	No edge before real costs means no edge after them. Real costs exceed our flat 20 bp, especially for a concentrated portfolio.	Backtest with spread- and size-dependent costs and capacity limits
The ESG floor did not cost return	**Does not survive**	Even here the floor itself adds nothing (all firms above the 80th percentile earn about the benchmark); the gain comes from a few names in one year (2022). In real markets a floor shrinks the universe, concentrates the portfolio and raises tracking error.	Random selections of the same size (done here: 11% to 19% match the rules); tracking error; other years
Cross-validation looks better than the test because the test years were a boom	**Survives as a lesson**	Models should be judged across regimes; random cross-validation mixes years and is optimistic.	Walk-forward evaluation over several market regimes

Closing recommendation (two lines): "We would not allocate on this evidence. Before relying on any ESG signal we would need several raters, dated vintages, delisting returns, realistic costs and an out-of-sample replication of the one finding."

6. Where every number in the memo is computed in the notebook

A memo result that the notebook does not reproduce earns no points. Notebook file: group_project_AS2026/notebook/Group_Work_AS_2026_solution.ipynb (Colab badge at the top). Cells are headed "(a) Step n" and so on.

Memo number	Notebook cell
28 firms, market cap 213,205 to 14,108, figure a1	(a) Step 2
300 duplicates, 9,002 to 9,182	(a) Step 3
Analyst_Outlook 0.68, test error 37.59 vs 29.14	(a) Step 4
Survivorship 4.62% vs 5.18%, 389 exits, 302 dropped firm-years	(a) Step 5
Restated: 459 firms, 0.010 vs 0.056, 86% overlap	(a) Step 6
Missing 13.2%, Europe 6.3%; scores at 0 or 100: 213 rows	(a) Step 7
Panel counts (10,410 rows, 30 features, 6,473 train, 2,709 test)	(a) Step 8
Feature table with dates	(a) Step 9
Audit table with effects	(a) Step 10
Model table, tuning grid, figures b1 and b2	(b) Steps 1, 2 and 3
Printed reading of (b): within-year check, rank correlation by year	(b) Step 5
Fold-by-fold CV (linear beats baseline in 5 of 5 folds), validation in time order	(b) Step 6
Region by period table, figure c1	(c) Steps 2 and 3
Restated vs point-in-time, joint regression	(c) Step 4
Europe in detail, year by year, years-as-unit p, figure c2	(c) Step 5
One-sentence conclusion (c)	(c) Step 6
Backtest table, robustness, 2022 to 2024 only	(d) Step 2
Bootstrap intervals	(d) Step 3
Cumulative return figure d1	(d) Step 4

Memo number	Notebook cell
Random selections, floor without forecast, without 2022, survivors 2018 to 2024, with 2025, block bootstrap	(d) Step 6
results.json (group name, members, all numbers)	Final cell at the end of the notebook

7. Slides: 6 slides for 8 minutes (plus about 2 minutes of questions)

The lecturer asked for one slide that explains the audit. The slide count and software are free, but 8 minutes is tight: six slides, about 1 to 1.5 minutes each. Every slide carries the dataset citation in a small footer line. Figures can be taken from results_sheets/figures/ (labelled axes, units).

Slide	Title and content	Time	Say (key sentences)
1	**Question and data.** The question; "all data simulated"; five files, 1,376 firms, 2012 to 2025; citation and licence	1:00	"Can ESG scores, or models built on them, earn an investor a premium? We tested this on a simulated panel, so the exercise tests our method, not the market."
2	**What we found in the data and what we did (the audit slide).** Compact table of 5 to 7 problems (problem, fix, effect); pipeline strip: 1 July rule, split 2015-2021 / 2022-2024, 30 dated features; figure a1 optional	1:30	"Seven problems. The most dangerous is a column that carries next year's return: a model using it cuts the test error from 37.6 to 29.1, but no investor had it on 1 July, so we excluded it."
3	**No model beats the naive forecast.** The four-row table or figure b1 (left); line: test years +21% boom, rank correlation about -0.02	1:30	"The models learn something in training but nothing that carries over. Complexity does not pay off."
4	**ESG paid in one place: Europe from 2021.** Figure c1 or the 14-cell table with Europe from 2021 highlighted; line: 1 of 14 cells, four years, restated scores look stronger	1:15	"+3.1 points per standard deviation, adjusted p 0.018; with years as the unit p is about 0.09. One cell, four years: plausible, not proven. Restated scores behave as if they contain hindsight and look stronger."
5	**Backtest with an ESG floor.** Figure d1 (left) or the four-row table; line: Sharpe difference +0.11, interval -0.16 to +0.38	1:30	"The floor did not cost return here, but the floor itself adds nothing: the gain is a few names in one year, 2022. The edge over the benchmark is not distinguishable from noise."
6	**What would change with real data.** The four named issues (raters disagree, point-in-time, survivorship, costs) and the survive / does not survive split; closing recommendation	1:15	"The method survives; the specific findings do not. We would not allocate on this evidence."

- **Rehearse twice with a timer** (Sat 5 and Mon 7 Dec). If you run over, cut slide 2 detail, never slide 6.
- Every group member must be able to answer questions on every slide: after each walkthrough, write three sentences in your own words per task.

8. Questions the lecturer may ask (and short answers)

Question	Short answer
Why divide by 1,000 and not drop the firms?	The environmental columns look normal for their size, so only revenue and market cap are wrong. Dropping loses 28 firms for nothing; the lecturer accepted any consistent, explained method.
How do you know Analyst_Outlook leaks?	Rank correlation 0.68 with the next 12 months and about 0 with the two earlier windows, and a test error that falls from 37.6 to 29.1 when it is used. We say it behaves as if it were stamped with future information; the data cannot show how it arose.
Why is the test error higher than the cross-validation error?	The test years were a boom (+21.6% against +0.9%), and cross-validation mixes years. Even after removing year averages the models show no skill.

Question	Short answer
Why linear regression?	Lowest cross-validation error, but a tie within 0.05 points; it has nothing to tune. We followed the rule set in advance and report gradient boosting as a labelled check.
Is the Europe result real?	It is the only cell that survives a multiple-testing correction with errors clustered by company, but it rests on four years (p about 0.09 with years as the unit) and is an association, not a tradable return.
Why Holm?	14 cells were examined. A false hit is likely by chance, so we raise the bar; only Europe from 2021 clears it.
Why is the restated version bigger?	What the restatement added correlates with the next return (rank correlation 0.16): it behaves as if it contains hindsight.
Did the ESG floor cost anything?	Not return in this sample. It cost concentration (19 names at the 80th percentile), volatility and trading. The gain is not an ESG effect: all firms above the floor earn about the benchmark, and without 2022 every rule trails it.
Why leave out formation year 2025?	It has only six months of returns. Including it lowers every rule and the benchmark by 1.6 to 2.3 points a year and keeps the ranking.
Is the benchmark comparison fair?	Equal weight on the full panel including firms that left, no cost, as in the starter; using survivors only would raise it from 8.77% to 9.22% a year over 2018 to 2024.
How realistic are 20 bp?	A simplification. Real costs depend on spread, size and market conditions, so net results would be lower.
Could the notebook be rerun?	Yes: Colab, Runtime, Restart session and run all; a few minutes; fixed seeds and fixed cross-validation folds.
What did you use AI for?	AI tools are encouraged by the lecturer. We used them for coding and reviewing; every group member can explain every cell. Keep a short log.

9. Style and quality checklist (the lecturer's bar)

- **Check every claim before writing it.** Only write "beyond X" or "controlling for Y" if the notebook ran that comparison. If unsure, ask Fabian.
- **Translate every number into what it means for an investor** (for example: an error of 37 points means the forecast cannot size positions).
- **Be precise with magnitude words** and check them against the numbers: "slightly", "much", "unusually". Avoid "unusually strong" for the test years (2018 was as strong); say "all three test years average about +21%".
- **Say "associated with"**, not "earned" or "caused", for the Europe result; say "behaves as if" for the restated scores and Analyst_Outlook.
- **State limitations plainly:** four years, 14 cells, one regime, 84 months, small tuning grids, flat costs.
- **Every metric next to its baseline; every figure with labelled axes and units; every feature dated.**
- **Keep numbers consistent:** one decimal convention (for example returns with one decimal, Sharpe ratios with two); use the numbers on the results sheets, not rounded versions from memory.
- **Write for an investment committee:** short sentences, no code names (say "Environmental score", not Environmental_Score), explain RMSE once as "typical forecast error".

10. Peer evaluation and open items

- **Peer evaluation:** every student rates the contribution of every group member (including themselves) in the Moodle form, by the memo deadline (Fri 27 Nov, 23:59). Where ratings and the presentation show clearly unequal contributions, the group score is adjusted for the individuals concerned.
- **To agree as a group:** the role wording for the memo (section 6), who presents which slide, and the rehearsal times.

- **Waiting on the lecturers:** the exact notebook deadline (will be close to the hand-in date in November) and confirmation that the talk is 8 minutes.
- **Fabian will tell you** if any number in the notebook changes; the results sheets then need a quick check.

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Team briefing for the memo and the slides

Rules that decide points, roles and deadlines

Item	Details
Institution	Bern University of Applied Sciences (BFH)
Course	EAIF: AI for Finance, Autumn Semester 2026
Group	Group_8 (Beyeler Fabian, Lussi Laurin, Griessen Jan)
Document type	Briefing for the writing lead and the slides
Key dates	Coaching Tue 24 Nov 2026; memo, notebook, results.json and peer evaluation Fri 27 Nov 2026, 23:59; slides Mon 7 Dec 2026, 23:59; presentation Tue 8 Dec 2026

> **Purpose.** A two-page orientation: what the assignment is, what to read first, the rules that decide points, the memo order and the deadlines.

What this is

A graded group assignment worth 100 points, one score for the group. We get simulated data about 1,376 companies (ESG scores and stock returns) and have to check it for errors, forecast returns, test whether good environmental scores paid, and test a trading strategy. **You do not need to code.** The notebook is built by the notebook owner, with AI help. You turn its results into the memo and the slides. Everyone must be able to explain what the notebook does, because the peer evaluation and the presentation can change individual scores.

Read first (about 45 minutes)

- **Coaching_Explainer_AS_2026.pdf**: the big picture, the four tasks, sections 1 to 5 and the glossary. The key idea is *leakage*: using information that the investor did not have at the time.
- **Group_assignment_AS_2026_transcription.md** (the assignment text): tasks (a) to (e), grading table and rules.
- Both are in the repo folder group_project_AS2026/docs/. Results sheets will appear in group_project_AS2026/results_sheets/.

Rules that decide points

- **Every number appears next to the baseline it must beat** (the naive average forecast for models, the equal-weight benchmark for the backtest).
- **Every figure has labelled axes with units.** Every feature is dated (the date from which it is known).
- **Say that the data is simulated** and cite it with its licence: "Simulated ESG and stock-return dataset, EAIF course, Autumn Semester 2026, CC BY 4.0", https://github.com/umatter/EDFB/tree/main/data/esg2026. Put it in the memo and on the slides.
- **Memo: 3 pages including tables and figures** (the lecturer confirmed), plus an appendix of any length. Group name and members on page 1.
- **No overclaiming.** A result like "the model does not beat the baseline" is accepted and graded as such. Write a claim only after the number behind it was checked.
- **Explain what each number means for an investor**, not only the number. Use size words that match the numbers.

Memo order (Word, 3 pages)

- 1. The question. 2. The data (simulated, licence). 3. The audit and the pipeline: features, the date from which each is known, the split.
- 4. Results of tasks (b), (c), (d), each metric next to its baseline.
- 5. A section titled **"What would change with real data"**: what remains unsolved after our audit (ESG scores are proprietary and raters disagree, survivorship is harder to see, trading costs are not constant, point-in-time availability) and, the larger part, which of our conclusions would survive and which would not, and why. Worth 15 points.
- 6. Group members and their roles.

Teammate B: writing lead

By	Deliverable
Sun 8 Nov	Memo skeleton in Word with the six headings. Data section with the simulated-data statement and licence line.
Wed 11 Nov	Audit paragraph (uses results sheet (a)).
Sun 15 Nov	Pipeline and model sections. Outline of "What would change with real data".
Sun 22 Nov	Complete memo draft (results sections from sheets (c) and (d)).
Tue 24 Nov	Coaching with the draft.
Thu 26 Nov	Final memo as PDF. Peer evaluation filled in. Submit evening.

Teammate C: analysis lead and slides

The talk is **8 minutes plus about 2 minutes of questions**, so plan 6 slides, about 1 to 1.5 minutes each. The lecturer asked for one slide that explains the audit: the problems found and how we treated them.

Slide	Content	Minutes
1	Question and data (state: simulated, licence)	1:00
2	Audit and pipeline: problems found, how treated	1:30
3	Model comparison: linear vs ensemble vs baseline	1:30
4	Where does ESG pay: region and period table	1:15
5	Backtest with ESG floor: table and cumulative return figure	1:30
6	What would change with real data	1:15

By	Deliverable
Sun 8 Nov	Slide skeleton in PowerPoint (6 slides, headings only).
Wed 11 Nov	Audit slide draft.
Sun 15 Nov	Model comparison slide.
Sun 22 Nov	Complete draft deck (also tables and figures with labelled axes and units).
Tue 24 Nov	Draft deck shown at the coaching.
Sat 5 and Mon 7 Dec	Two timed rehearsals. Final PDF by Mon 7 Dec noon (deadline 23:59).
Tue 8 Dec	Presentation.

What you get from the notebook owner

- After each task: a **one-page results sheet** (the numbers, each next to its baseline, with suggested wording) and a **plain-language guide** to what the code did.
- A **30-minute walkthrough** on Sun 8, Sun 15 and Sun 22 Nov. After each one, write three sentences in your own words about what was done and why. If you can do that, you can answer questions on it.
- If a number looks odd, ask before you write it down.

Everyone

- All three of us must be able to answer questions on every slide and every part of the notebook, in plain terms.
- Each student fills in the **peer evaluation** on Moodle by the memo deadline.
- Coaching with the lecturer: **Tue 24 Nov**. Bring the draft memo and draft deck.

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Task (a): results sheet and guide

Pipeline and data audit

Item	Details
Institution	Bern University of Applied Sciences (BFH)
Course	EAIF: AI for Finance, Autumn Semester 2026
Group	Group_8 (Beyeler Fabian, Lussi Laurin, Griessen Jan)
Document type	Results sheet and plain-language guide for task (a)
Source of numbers	Executed solution notebook, local run of 4 November 2026 (a fresh Colab run follows before submission)

> **Purpose.** The audit table with its numbers, what each problem means for an investor, suggested memo and slide text, a plain-language guide to the code, and likely questions.

1. The result in four sentences

We checked all five data files before building any model and found **seven problems**, which are treated in the notebook. Three of them change numbers a reader would notice: a unit error made 28 firms look about 1,000 times bigger than they are, duplicated return months silently removed 181 firm-years from the model data, and the column Analyst_Outlook carries information from the future. We excluded that column, because no investor could have had it on 1 July. Then we built the **formation panel**: one row per company and formation year, with 30 features, six of them new, each with the date from which it is known.

2. The audit table (numbers from the notebook)

Problem	How found	What we did	Effect on one number
Unit error: 28 firms (302 firm-years) have revenue and market cap about 1,000 times too large	Carbon, water and energy scale with revenue; for these firms they sit about 1,000 times below that relation	Divided revenue and market cap by 1,000	Mean market cap 213,205 to 14,108 USD million. These firms were 100% of the 100 largest firm-years by market cap before, 0% after.
Duplicate returns: 300 company-months appear twice (values differ by at most 0.01 points)	Repeated (Company_ID, Month) keys	Kept the first row	Firm-years with a full 12 holding months: 9,002 to 9,182. 181 firm-years had 13 months and had been dropped.
Analyst_Outlook carries future information	Rank correlation with the return in three 12-month windows	Excluded from all models	Rank correlation with the next 12-month return 0.68 (other windows -0.01 and 0.01). Random forest test error 37.59 without it, 29.14 with it, baseline 37.20.
Survivorship: the universe file lists 987 survivors out of 1,376 firms	389 firms have a last fiscal year before 2025	Everything built on the full panel, exiting firms held until their last month	Equal-weight benchmark 4.62% a year on the full panel vs 5.18% on survivors only (Sharpe 0.213 vs 0.248), formation years 2015 to 2025. The model samples also drop 302 firm-years of exiting firms (fewer than 12 holding months).

Problem	How found	What we did	Effect on one number
Restated scores behave as if they contain hindsight	Compared with the point-in-time scores; changes correlate with later returns	Point-in-time scores everywhere; restated only for the comparison in task (c)	459 firms have every score changed (by 5.7 points on average). Within-year rank correlation of the Environmental score with the next excess return 0.010 (point-in-time) vs 0.056 (restated).
Missing environmental data: carbon, water, energy missing together, not at random	Missing share by region and by score	Kept all rows; carbon intensity filled with the industry-year median plus a missing flag	13.2% of formation rows. Europe 6.3% missing vs 13.0% to 18.9% elsewhere. Mean Environmental score 47.2 (missing) vs 56.5. Dropping the carbon features changes the linear test error from 38.10 to 38.05.
Values at the bounds: scores at exactly 0 or 100 (65 Environmental, 72 Social, 77 Governance; 213 rows, 1.5%) and 8 returns at -95%	Counts at the minimum and maximum	Kept and documented	Removing the clipped training rows changes the linear test error from 38.10 to 38.00.

3. What the numbers mean for an investor

- **Unit error.** Left in, these 28 firms would be treated as the largest firms in the data and would distort every feature that depends on size. The mean market cap would be overstated by a factor of about 15.
- **Duplicates.** One extra month changes a firm's 12-month return, and a firm with 13 months does not pass the "exactly 12 months" rule, so it was silently left out of the model data.
- **Analyst_Outlook.** A model using it looks far better than the baseline (29.1 against 37.2), but only because the column contains the outcome. An investor on 1 July could not have had it. The inference that it carries future information rests on the pattern of correlations: it is related to the next 12 months and to nothing else.
- **Survivorship.** A test on survivors alone would make an equal-weight portfolio look about 0.6 percentage points a year better than it was (formation years 2015 to 2025; about 0.46 points over the 2018 to 2024 backtest window of task d), because the firms that failed are missing.
- **Restated scores.** The restated Environmental score is more strongly related to later returns than the score investors actually saw (0.056 against 0.010). Using it to "test" a strategy would suggest more of an ESG premium than could have been earned.
- **Missing data.** Dropping the rows with gaps would remove 13.2% of the sample and tilt it towards firms with higher Environmental scores.

4. The formation panel and the features

Item	Number
Formation rows (one per company and formation year, 2015 to 2025)	10,410
Rows with a full 12 holding months (used to fit and test models)	9,182
Training rows (formation years up to 2021) / test rows (2022 to 2024)	6,473 / 2,709
Naive baseline: training mean 12-month return / test RMSE	0.94% / 37.20 percentage points
Mean 12-month return, training years / test years	0.9% / 21.6%

The test years were strong years for the data (mean 12-month return +21.6% against +0.9% in training). This is why a model that is trained on the earlier years looks poor on the test years. The starter notebook mentions a market downturn, but the data shows the opposite.

New feature	What it is	Known from
Env_Change	Environmental score minus the previous year's	Published 31 Mar to 31 May of Y+1, so before 1 July
Margin_Change	Profit margin minus the previous year's	Same
log_MC_to_Rev	Log of market cap divided by revenue (valuation relative to sales)	Same
log_Carbon_Int	Log of carbon emissions per USD million revenue; gaps filled with the industry-year median	Same
Carbon_Missing	1 if carbon, water and energy are missing	Same
Past_Vol_Pct	Standard deviation of the 12 monthly returns before formation	Returns through 30 June of the formation year

5. Suggested text (edit freely)

Memo, audit paragraph (about 110 words). Before modelling, we audited all five files and found seven problems. Revenue and market cap of 28 firms were about 1,000 times too large, probably recorded in USD thousand, so we divided them by 1,000. 300 company-months appeared twice in the returns file and we kept the first row. The column Analyst_Outlook is closely tied to the next 12-month return (rank correlation 0.68, about zero for other windows), so it cannot have been known on 1 July and we excluded it. The universe file holds survivors only, so we used the full panel. Restated ESG scores behave as if they contain hindsight, so we used the point-in-time scores. Missing environmental data is not random and was flagged, not dropped.

Slide (the audit slide the lecturer suggested). Title: "What we found in the data and what we did". A table with five rows (Problem, Fix, Effect): unit error, duplicates, Analyst_Outlook, survivorship, restated scores. One line below: "Seven problems found and treated; every feature is dated and known on 1 July." Speaking time about 1.5 minutes.

6. Plain-language guide: what the notebook did

- **Step 1, structure checks.** Looked for duplicate keys, gaps and publication dates outside the expected window. Only the returns file had duplicates.
- **Step 2, unit error.** Within an industry, a firm's carbon, water and energy use grow with its revenue. For 28 firms the revenue was about 1,000 times too large for that relationship, so we divided it (and market cap) by 1,000.
- **Step 3, duplicates.** Kept one row for each company-month that appeared twice.
- **Step 4, Analyst_Outlook.** Compared it with the return in three different 12-month windows. It matched only the window that is the thing we try to predict. Excluded.
- **Step 5, survivorship.** Compared a benchmark of all firms with a benchmark of survivors only.
- **Step 6, restated scores.** Compared both score files and checked how each relates to later returns.
- **Step 7, missing data and bounds.** Described the gaps and the values at 0, 100 and -95%.
- **Steps 8 to 10, audit table, panel, feature table.** Collected the numbers into one table, built the panel with the new features, and checked in code that no row is used before it was published.

7. Questions you may be asked

Question	Answer
Why divide by 1,000 instead of dropping the 28 firms?	Dropping them loses 28 firms for nothing. Their carbon, water and energy figures look normal for their size, so only revenue and market cap are wrong. The lecturer's feedback also allowed any consistent, explained method.

Question	Answer
How do you know it is a unit error?	The 28 firms show the same factor of 1,000 on revenue and market cap, and the environmental columns are normal. We cannot confirm it with the data owner, so we say "probably".
Why keep the first duplicate?	The two values differ by at most 0.01 percentage points, so keeping the first or averaging gives practically the same result.
How do you know Analyst_Outlook leaks?	It has a rank correlation of 0.68 with the next 12-month return and about zero with the two earlier windows. A genuine forecast would not be that good. Using it would cut the test error from 37.6 to 29.1.
Why not use the universe file?	It lists only the 987 firms that still exist. The 389 that left are missing, which makes returns look better than they were.
Why not drop rows with missing environmental data?	It would remove 13.2% of the sample and bias it towards higher scores, because missing rows have lower Environmental scores.

8. What to be careful about

- The unit fix is an inference, not confirmed. Say "probably".
- The correlations in this sheet are averaged over all regions and years. Task (c) tests the Environmental score region by region and period by period, and may show something different.
- Each effect was measured one problem at a time, on the data as it stood before the other fixes, so the effects do not add up.
- All numbers come from the notebook run of 4 November. If the notebook changes, the numbers must be checked again.

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Task (b): results sheet and guide

Linear regression vs tree ensembles

Item	Details
Institution	Bern University of Applied Sciences (BFH)
Course	EAIF: AI for Finance, Autumn Semester 2026
Group	Group_8 (Beyeler Fabian, Lussi Laurin, Griessen Jan)
Document type	Results sheet and plain-language guide for task (b)
Source of numbers	Executed solution notebook, local run of 4 November 2026 (a fresh Colab run follows before submission)

> **Purpose.** The model comparison next to the baseline, what the errors mean for an investor, suggested memo and slide text, a plain-language guide to the code, and likely questions.

1. The result in four sentences

We forecast each firm's total return over the next 12 months from 30 features that are known on 1 July, training on formation years 2015 to 2021 and testing on 2022 to 2024. **No model beats the naive forecast** (the training average) on the test years: the baseline's test error is 37.20 percentage points, against 38.10 for linear regression, 37.93 for a tuned random forest and 37.84 for tuned gradient boosting. Cross-validation selects linear regression, but the three models are within 0.05 points of each other, so the choice is effectively a tie. The extra complexity of the ensembles does not pay off out of sample, which the assignment accepts as a legitimate result.

2. The comparison table (every model next to the baseline)

Model	Train RMSE	CV RMSE (sd)	Test RMSE	Train R2	CV R2	Test R2	Test rank corr.
Naive mean (baseline)	29.16	29.16 (0.29)	**37.20**	0.00	0.00	-0.45	n/a
Linear regression	28.73	**28.88** (0.36)	38.10	0.03	0.02	-0.52	-0.02
Random forest (tuned)	28.11	28.92 (0.29)	37.93	0.07	0.02	-0.51	-0.03
Gradient boosting (tuned)	28.28	28.90 (0.32)	37.84	0.06	0.02	-0.50	-0.02

RMSE = typical size of the forecast error in percentage points of 12-month return (lower is better). CV = cross-validation inside the training years, folds grouped by company. Test rank corr. = average over the three test years of the rank correlation between forecast and outcome. Settings chosen by CV: random forest 150 trees, depth 6, minimum leaf 60; gradient boosting 100 trees, learning rate 0.03, depth 3. Selected on CV: linear regression.

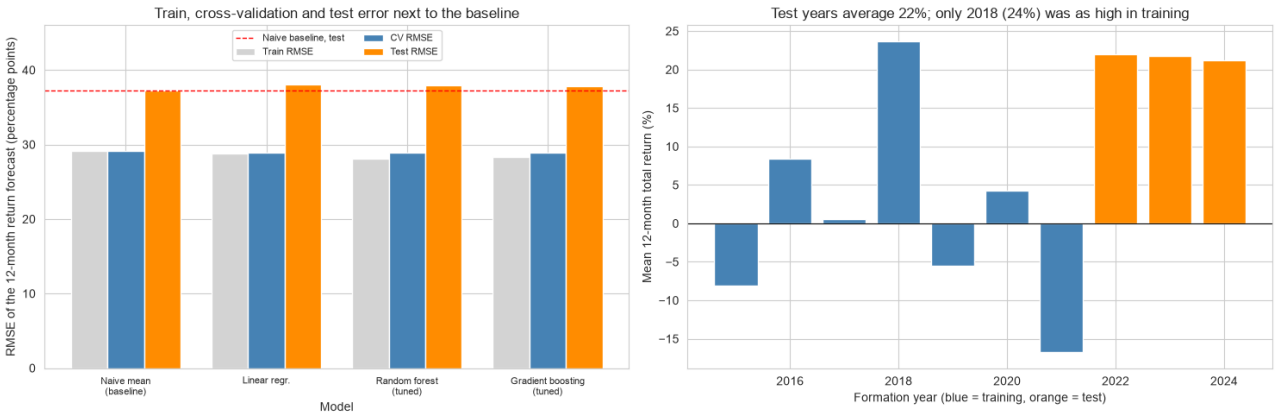


Figure b1 (notebook, (b) Step 3): left, train, CV and test error next to the baseline; right, the mean 12-month return by formation year.

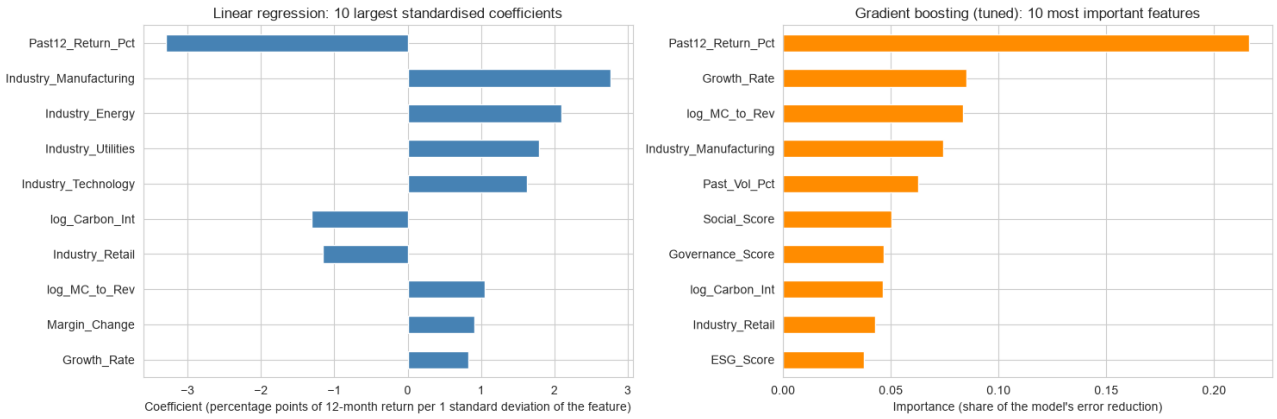


Figure b2: the 10 largest standardised coefficients of the linear model and the 10 most important features of gradient boosting.

3. What the numbers mean for an investor

- **The forecasts cannot be used.** An error of 37 to 38 points on 12-month returns that average between +1% and +22% is far too large to size or rank positions. Cross-validated R2 is about 2%: in the training years the models explain almost nothing.
- **R2 is negative for everyone, including the baseline (-0.45).** That is because R2 here is measured against the test years' own average, and the test years were a boom. Measured against the training-average forecast, the models are slightly worse than guessing the average (R2 -0.05 linear, -0.04 forest, -0.03 boosting).
- **Why test error (37 to 38) is so much higher than CV error (28.9).** The three test years all averaged about +21% (+22.0%, +21.7%, +21.2%), against +0.9% in the training years; only 2018 (+24%) was as high before. A forecast built on the training average is far too low. A naive forecast that already knew each test year's average would score 30.9. Cross-validation mixes years inside the training period and cannot see this shift. The starter notebook calls the test years a downturn; the data show the opposite.
- **It is not only the level shift.** After removing each year's average from outcome and forecast, the models still do not beat the zero forecast (30.91): linear 31.40, forest 31.31, boosting 31.23. And the rank correlation between forecast and outcome flips sign across years (linear +0.07 in 2022, -0.05 in 2023, -0.09 in 2024; average -0.02): no reliable ability to rank firms.
- **The cross-validation gain over the baseline is small but consistent, and it does not carry forward in time.** Linear regression's CV error is 28.88 against 29.16 for the baseline, a gain of 0.28, and it is lower in all 5 folds (by 0.07 to 0.47). But validated in time order inside the training years (train on earlier years, check on the next one, 2017 to 2021), no model beats the naive forecast (average error 30.46 for the baseline, 30.63 to 30.77 for the models), which matches the test result ((b) Step 6).

- **One feature drives the linear model's result.** Its largest coefficient (-3.3 points per standard deviation) is on the past 12-month return, which carries market-wide year-to-year variation learned from only 7 training years. Without that feature its test error is 37.69: better, but still above the baseline.

4. Suggested text (edit freely)

Memo (about 120 words). We forecast each firm's 12-month total return from 30 features known on 1 July, training on formation years 2015 to 2021 and testing on 2022 to 2024. Linear regression (selected on cross-validation), a tuned random forest and tuned gradient boosting were compared with the naive training-average forecast. None beat the baseline out of sample: test RMSE is 37.20 for the baseline and 38.10, 37.93 and 37.84 for the models. The three models are within 0.05 points in cross-validation, so the selection is effectively a tie. The test years were a uniform boom (average 12-month return +21.6%, against +0.9% in training), and even after removing year averages the models show no cross-sectional skill (rank correlation about -0.02). The extra complexity of the ensembles does not pay off out of sample.

Slide (model comparison, about 1.5 minutes). Title: "No model beats the naive forecast". The four-row table (baseline, linear, random forest, boosting; train, CV, test RMSE) or figure b1 (left). One line below: "Test years were a +21% boom; no cross-sectional skill (rank correlation -0.02)". Say: the models learn something in training, but nothing that carries to the test years.

5. Plain-language guide: what the notebook did

- **Step 1, setup and tuning.** Took the table of firm-years from task (a), split it by year (training up to 2021, test 2022 to 2024). Each ensemble has settings (tree depth, learning rate). We tried 4 settings for each and kept the best by cross-validation: cut the training data into 5 groups of companies, train on 4, check on the 5th, rotate. All rows of a company stay in one group, so a model cannot recognise a company it already saw.
- **Step 2, side-by-side table.** Fitted every model on the training years, measured train, CV and test error next to the naive forecast, and chose the model with the lowest CV error. The test years are used once, here.
- **Step 3, figures.** Errors next to the baseline, the yearly average returns, and which features matter.
- **Step 4, numbers for results.json.** Stored the baseline, linear, best-ensemble errors and the selected model.
- **Step 5, reading.** Computed the sentences above from the table (including the year-by-year rank correlation and the within-year check), so they update if the data change. Test errors reported here are diagnostics, not used to choose anything.
- **Step 6, two checks on the selection (training years only).** The error fold by fold (linear beats the baseline in every fold, but the best of the three models changes from fold to fold) and a validation in time order (no model beats the naive forecast).

6. Questions you may be asked

Question	Answer
Why linear regression, when boosting has the best test error?	The model is chosen on cross-validation only; the test years are used once, to report, not to choose. Choosing on the test result would flatter it. The three models are in any case within 0.05 CV points.
Why is test R2 negative?	R2 compares with the test years' own average. All three test years were about +21% and the models predict near the training average (about +1%), so every forecast, including the baseline (-0.45), looks bad on that scale.
Are the models useless?	They show no skill out of sample here. That is common: 12-month stock returns are very hard to forecast. The assignment accepts this result.

Question	Answer
What did you tune in linear regression?	Nothing: ordinary linear regression has no setting to tune, so it enters as it is. Only the two ensembles were tuned, on 4 settings each.
Why group the folds by company?	Otherwise the same company would sit in training and in the check set, and the model could recognise it instead of forecasting. The folds still mix years, which makes CV a little optimistic.
Could a different year split change the answer?	Possibly. The test years are one regime (a boom). We report that plainly and do not claim more than 'no evidence of skill'.

7. What to be careful about

- The selection of linear regression is a near-tie: the three models are within 0.05 CV points, and the best of them changes from fold to fold (linear regression wins 3 of 5 folds; (b) Step 6). Never say it won clearly.
- The test set covers three years of one regime; "no model beats the baseline" rests on that.
- The tuning grids are small (4 settings per ensemble, 100 to 150 trees), and the folds mix years.
- Rows with fewer than 12 holding months (302 firm-years of exiting firms) are not in the model samples.
- All numbers come from the run of 4 November; the notebook's printed sentences are recomputed on every run, this sheet is not.

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Task (c): results sheet and guide

Where does ESG pay?

Item	Details
Institution	Bern University of Applied Sciences (BFH)
Course	EAIF: AI for Finance, Autumn Semester 2026
Group	Group_8 (Beyeler Fabian, Lussi Laurin, Griessen Jan)
Document type	Results sheet and plain-language guide for task (c)
Source of numbers	Executed solution notebook, local run of 4 November 2026 (a fresh Colab run follows before submission)

> **Purpose.** The region by period table (point-in-time and restated), what the one significant cell means and how firm it is, suggested memo and slide text, and likely questions.

1. The result in four sentences

We related the previous fiscal year's **Environmental score** (standardised within each year) to the **next 12 months' excess return**, region by region and for formation years before 2021 and from 2021, comparing each firm with same-industry firms of the same year. Pooled over all regions there is nothing. Region by region, exactly **one cell** stands out: **Europe, from 2021**, where a firm one standard deviation (about 17 points) higher on the score was associated with about **+3.1 percentage points** more 12-month excess return (95% interval 1.2 to 5.0; Holm-adjusted p = 0.018). No other region-period cell is distinguishable from zero. The evidence rests on only four formation years: with years as the unit instead of firms, p is about 0.09, so we call it a plausible premium, not a proven one. The restated scores, which behave as if they contain hindsight, give higher estimates everywhere and more significant cells before the correction; after it they, too, keep only Europe from 2021 (+4.88).

2. The table (percentage points of 12-month excess return per 1 standard deviation of the score; standard error in brackets)

Region	Point-in-time, before 2021	Point-in-time, from 2021	Restated, before 2021	Restated, from 2021
Africa	-0.07 (1.45)	+0.75 (1.43)	+0.93 (1.38)	+2.40 (1.25)
Asia	+0.15 (0.68)	+0.39 (0.96)	+1.06 (0.67)	+1.95 (0.94) *
Europe	+0.47 (0.65)	**+3.08 (0.96)** **	+1.59 (0.68) *	+4.88 (0.90) **
Latin America	-1.19 (1.08)	-2.94 (1.66)	-0.05 (1.14)	-1.33 (1.77)
Middle East	-0.92 (1.08)	+0.28 (1.57)	+0.43 (1.11)	+2.75 (1.55)
North America	-0.40 (0.72)	-0.61 (1.19)	+0.93 (0.71)	+0.99 (1.21)
Oceania	-0.24 (0.88)	-0.80 (1.33)	+0.73 (0.87)	+0.79 (1.31)
All regions (reference)	-0.12 (0.28)	+0.63 (0.43)	+0.87 (0.28) *	+2.13 (0.42) *

* = p < 0.05 for that cell alone; ** = still p < 0.05 after the Holm correction across the 14 region-period cells (the 'All regions' rows are reference only). Standard errors are clustered by company; controls are industry-by-formation-year effects. Cell sizes (firm-years): Europe 1,659 before and 814 from 2021; other regions 667 to 1,686 before and 340 to 817 from 2021. Europe, from 2021, point-in-time: 95% interval 1.20 to 4.96, p = 0.0013, Holm 0.018, 814 firm-years, 223 firms.

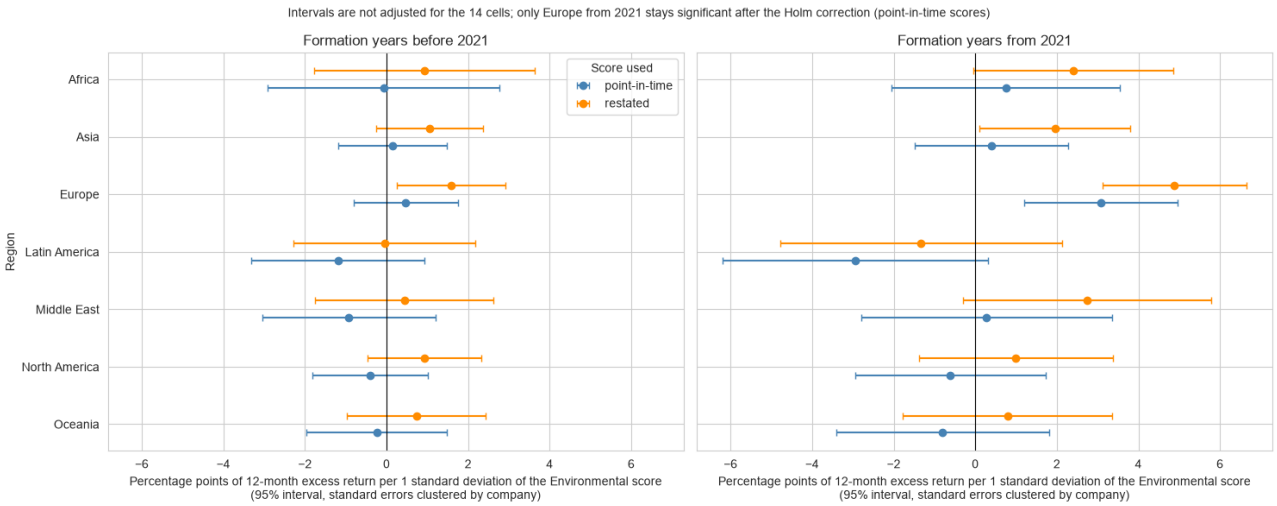


Figure c1 (notebook, (c) Step 3): estimates and 95% intervals by region and period, point-in-time (blue) and restated (orange). The intervals are not adjusted for the 14 cells.

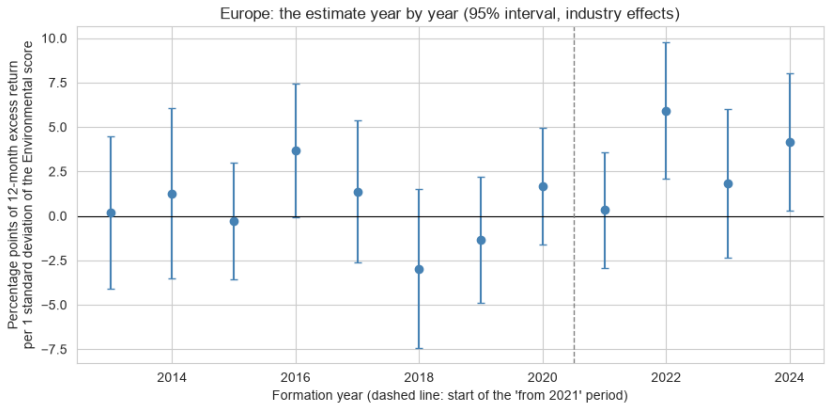


Figure c2: the Europe estimate year by year (95% interval). The premium appears from 2022: 2021 is +0.35.

3. What the numbers mean for an investor

- **Where could an investor have earned a premium?** Only among European firms from 2021: a European firm about 17 points above its year's average Environmental score was associated with about 3 points more 12-month excess return than industry peers of the same year. Everywhere else, and before 2021, the data show no relation that can be told apart from zero.
- **This is an association, not a return.** The table does not show that a portfolio built on the score would have earned this after costs; task (d) tests a ranking rule, not this region-specific effect.
- **How firm is it?** The firm-clustered p-value (0.0013) treats firms as independent. With years as the unit (Fama-MacBeth over the four yearly slopes +0.35, +5.93, +1.81, +4.18) the p-value is about 0.09. We examined 14 cells and 12 yearly estimates, so we call it a plausible premium, not a proven one.
- **Is it different from before 2021?** Europe before 2021 is +0.47 (se 0.65); the extra effect from 2021 is +2.61 (se 1.26, p = 0.038), an exploratory test run after the cell was singled out.
- **Why pooling hides it.** Pooled over all regions the estimate is -0.12 before 2021 and +0.63 from 2021: one region cannot be seen in a global ranking.

The restated scores: why the estimates differ

- The restated file gives higher coefficients in every region and period (by 0.9 to 2.5 points) and, before the Holm correction, makes more cells look significant, for example Europe before 2021 (+1.59) and Asia from 2021 (+1.95), where the point-in-time scores show nothing. After the correction only Europe from 2021 remains (+4.88).

- The reason: the restatement behaves as if it contains hindsight. What it added to the score is related to the return that followed (within-year rank correlation 0.16; in a joint regression +5.3 points per standard deviation of the restatement). An investor on 1 July would have seen only the point-in-time score, so the restated version overstates what could have been earned and points to premiums that did not exist at the time.

4. Suggested text (edit freely)

Memo (about 130 words). We estimated, for each region and for formation years before and from 2021, the relation between the previous fiscal year's Environmental score (standardised within year) and the next 12 months' excess return, comparing each firm with same-industry firms of the same year (standard errors clustered by company, Holm correction across the 14 cells). Pooled over regions the relation is indistinguishable from zero. An investor could have earned an ESG premium only in Europe from 2021: about +3.1 percentage points per standard deviation of the score (Holm-adjusted p = 0.018), found in 1 of 14 cells and resting on four formation years (p about 0.09 with years as the unit). Restated scores, which behave as if they contain hindsight, give higher estimates everywhere and more significant cells before the correction.

Slide (where ESG pays, about 1.25 minutes). Title: "ESG paid in one place: Europe from 2021". Figure c1 (or a table of the 14 cells with Europe from 2021 highlighted). One line: "1 of 14 cells; +3.1 points per standard deviation; restated scores look stronger (hindsight)".

5. Plain-language guide: what the notebook did

- **Step 1, sample.** One row per firm and year, 2013 to 2024, with a full 12 months of returns. The score is standardised within each year (average 0, spread 1). The outcome is the excess return: total return minus what a safe account paid.
- **Step 2, one regression per cell.** For each region and period: how much does the next-year excess return change when the score is one spread higher, comparing only firms in the same industry in the same year? We compute the uncertainty with the firm-clustered method (a firm's years are not independent evidence) and apply the Holm correction, which raises the bar because looking at 14 cells can produce a lucky hit by chance. The same is done with the restated scores.
- **Step 3, figure** of all estimates with intervals.
- **Step 4, why restated differs.** Compared the two score versions and showed that what the restatement added correlates with later returns.
- **Step 5, the Europe cell in detail:** a test against Europe before 2021, year-by-year estimates, and the years-as-unit test (p about 0.09). Marked exploratory.
- **Step 6, numbers and the one-sentence conclusion** for results.json.

6. Questions you may be asked

Question	Answer
Is this causal? Does a higher score cause higher returns?	No. It is a cross-sectional association within industry and year. We make no causal claim.
Why the Holm correction?	We look at 14 cells. At the 5% level one false hit is expected about every 20 looks, so we raise the bar. Only Europe from 2021 clears it (adjusted p = 0.018).
Why cluster by company?	The same company appears in many years and its years resemble each other; counting them as independent would make us overconfident.
Why does Europe from 2021 start in 2022?	The 2021 formation year is +0.35. The cell averages 2021 to 2024; the premium shows in 2022 (+5.9), 2023 (+1.8, wide) and 2024 (+4.2).

Question	Answer
Why is the restated version bigger?	It behaves as if it contains hindsight: what it added correlates with the return that followed. It overstates what an investor could have known.
Does the result depend on how you measure uncertainty?	Yes. Holm-adjusted p = 0.018 holds with errors clustered by company. With the four formation years as the unit (Fama-MacBeth) p is about 0.09. We report both, which is why we say plausible, not proven.
Is 'Europe from 2021' the planted relationship?	We cannot know. It is the only cell that survives a multiple-testing correction. In real data a single cell out of 14 would need replication before anyone relied on it.

7. What to be careful about

- Say "associated with", never "earned" without the caveat that it is not a tradable return.
- Four formation years: do not call it proven. The intervals in the figure are unadjusted for the 14 cells.
- The cutoff "from 2021" is by formation year (agreed in the coaching).
- The 3.1 points are per standard deviation of the score (about 17 points), not per score point.
- Numbers come from the run of 4 November.

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Task (d): results sheet and guide

Backtest with costs and an ESG floor

Item	Details
Institution	Bern University of Applied Sciences (BFH)
Course	EAIF: AI for Finance, Autumn Semester 2026
Group	Group_8 (Beyeler Fabian, Lussi Laurin, Griessen Jan)
Document type	Results sheet and plain-language guide for task (d)
Source of numbers	Executed solution notebook, local run of 4 November 2026 (a fresh Colab run follows before submission)

> **Purpose.** The backtest table next to the benchmark, the bootstrap, what the floor costs, suggested memo and slide text, and likely questions.

1. The result in four sentences

Each 1 July we ranked firms on the forecast of the model selected in task (b) (linear regression, refitted every year on earlier years only), held the top decile for twelve months and charged 20 basis points per unit of one-way turnover, on the full panel including firms that left (formation years 2018 to 2024, 84 months). The rule earned **9.4% a year against 8.8%** for the equal-weight benchmark (Sharpe ratio 0.46 against 0.42). Restricting the pool to firms with an Environmental score at or above the 50th and the 80th percentile gave 9.6% and 11.4%: **in this sample the floor did not cost return**, it cost concentration (about 19 names at the 80th percentile) and trading. The best rule's Sharpe advantage over the benchmark (+0.11) is **not distinguishable from noise** (95% interval -0.16 to +0.38), and the floor result is **not an ESG effect**: holding every firm above the 80th percentile earns about the benchmark (8.9%), and the gap comes from a few names picked in formation year 2022.

2. The table (July 2018 to June 2025, after costs, full panel including firms that left)

Rule	Return (% a year)	Volatility (% a year)	Sharpe ratio	One-way turnover	Names held
Linear regression, no floor	9.39	16.80	0.457	0.70	94
Linear regression, floor 50th percentile	9.59	17.08	0.464	0.76	47
Linear regression, floor 80th percentile	**11.39**	18.38	**0.533**	0.82	19
Equal-weight benchmark (all firms)	**8.77**	17.12	**0.419**	0.17 (no cost)	945

Cost convention (from the starter): the replaced share of the portfolio is sold and bought, 20 bp each way, so 40 bp on the replaced share, charged in the first month of each holding year. Cost drag per year: 0.28 points (no floor), 0.31 (50th), 0.33 (80th). The benchmark is charged no cost, as in the starter. Floors: Environmental score at or above the 50th or 80th percentile of that formation year; the top decile is 10% of the restricted pool.

Sharpe difference to the benchmark (95% interval, 5,000 resamples of the 84 months)	Difference	Lower	Upper
No floor minus benchmark	+0.038	-0.159	+0.226
Floor 50th minus benchmark	+0.045	-0.176	+0.255

Sharpe difference to the benchmark (95% interval, 5,000 resamples of the 84 months)	Difference	Lower	Upper
Floor 80th minus benchmark (best rule)	**+0.114**	**-0.162**	**+0.381**

What drives the result ((d) Step 6)	No floor	Floor 50th	Floor 80th	Benchmark
The rule, return % a year (from the table above)	9.39	9.59	11.39	8.77
Every firm above the floor at equal weight, no forecast	n/a	8.75	8.88	8.77
The rule without formation year 2022 (72 months)	6.05	6.20	6.58	6.70
Holding year of formation 2022 alone (%)	31.7	32.4	45.2	22.0
Share of 1,000 random selections of the same size that do at least as well	16%	19%	11%	n/a

Reading: the floor itself adds nothing (row 2 is about the benchmark), the whole edge of every rule comes from formation year 2022 (row 3), and a random selection of the same size would match the rule in 11% to 19% of cases (row 5).

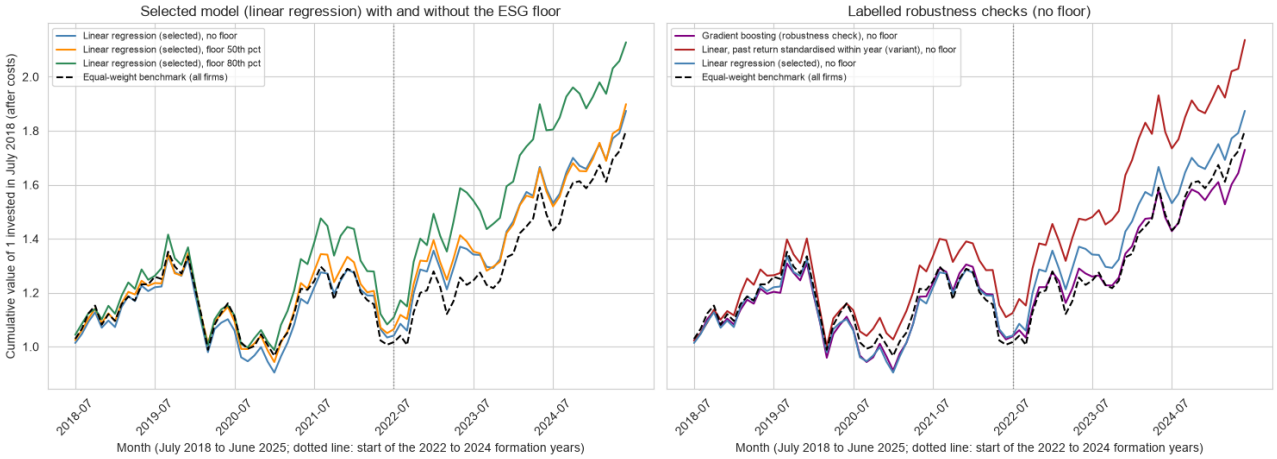


Figure d1 (notebook, (d) Step 4): cumulative value of 1 invested in July 2018 after costs. Left: the selected model with and without the floor against the benchmark. Right: the labelled robustness checks.

Only the clean out-of-sample years (formation 2022 to 2024, 36 months)	Return	Volatility	Sharpe
Linear regression, no floor	21.92	14.70	1.111
Linear regression, floor 50th	21.83	14.92	1.092
Linear regression, floor 80th	25.28	15.74	1.222
Equal-weight benchmark	21.37	15.17	1.052

Labelled robustness checks (not chosen on cross-validation)	Return	Volatility	Sharpe	Turnover
Gradient boosting, no floor	8.14	16.71	0.390	0.78
Gradient boosting, floor 50th / floor 80th	9.25 / 9.56	16.90 / 17.71	0.448 / 0.452	0.82 / 0.86
Linear, past return standardised within year, no floor	11.46	16.89	0.568	0.55
Same variant, floor 50th / floor 80th	10.83 / 10.24	16.92 / 18.08	0.534 / 0.481	0.60 / 0.69

3. What the numbers mean for an investor

- **No convincing edge.** The forecast has almost no ranking skill: the rank correlation between forecast and realised return averages +0.01 over 2018 to 2024 (+0.03 in 2018 to 2021, -0.02 in 2022 to 2024, and it changes sign across

years). The strategy holds about 94 names against about 945 for the benchmark, and 16% of random selections of the same size from the same pool do at least as well: a gap of 0.6 points a year is within what an unskilled selection of names produces.

- **What the floor costs.** Not return in this sample. The 50th-percentile floor earned +0.2 points against no floor and the 80th +2.0 points. What it cost: fewer names (47 and 19 instead of 94), more volatility (17.1% and 18.4% against 16.8%) and more trading (turnover 0.76 and 0.82 against 0.70, a cost drag of 0.31 and 0.33 points a year against 0.28). The 80th-percentile result is **not an ESG effect**: holding every firm above the 80th percentile at equal weight earns 8.9%, about the benchmark's 8.8%, so the floor itself adds nothing. The gap comes from the about 19 names the model picked, mostly in one year (formation 2022: +45% against +22%). Without formation year 2022 every rule trails the benchmark (6.0%, 6.2% and 6.6% against 6.7%).
- **Is the difference to the benchmark noise?** Yes, for all three rules the interval contains zero. The best rule's Sharpe difference is +0.11 (interval -0.16 to +0.38). Two cautions: picking the best of three rules after seeing the results flatters it, and 84 months are few. Resampling blocks of 3, 6 or 12 months instead of single months gives intervals that are no wider (12-month blocks: -0.07 to +0.31).
- **The clean out-of-sample years** (July 2022 to June 2025, a strong market) tell the same story: 21.9% against 21.4% without a floor; the Sharpe ratios 1.11 and 1.05 are close.
- **Robustness.** Gradient boosting earned 8.1% without a floor (below the benchmark). A linear variant with the past return standardised within year earned 11.5% (Sharpe 0.57), the highest no-floor result, but it was not the model selected in task (b), and choosing it now because it looks best would be selecting on the test period.

4. Suggested text (edit freely)

Memo (about 140 words). Each 1 July we ranked firms on the forecast of the model selected in (b) (linear regression, refitted each year on earlier years only), held the top decile for twelve months and charged 20 bp per unit of one-way turnover, on the full panel including firms that left (formation years 2018 to 2024). The rule earned 9.4% a year against 8.8% for the equal-weight benchmark (Sharpe 0.46 against 0.42). Floors at the 50th and 80th percentile of the Environmental score gave 9.6% and 11.4%: the floor did not cost return here, only concentration (about 19 names at the 80th percentile) and trading. This is not an ESG effect: all firms above the 80th percentile at equal weight earn 8.9%, and without formation year 2022 every rule trails the benchmark. The best rule's Sharpe advantage (+0.11, 95% interval -0.16 to +0.38) is not distinguishable from noise.

Slide (backtest with the floor, about 1.5 minutes). Title: "A backtest with an ESG floor: no edge that beats noise". Figure d1 (left) or the four-row table. One line: "Sharpe difference to the benchmark +0.11, interval -0.16 to +0.38; the floor adds nothing by itself; the gap is one year (2022)".

5. Plain-language guide: what the notebook did

- **Step 1, forecasts.** For every formation year from 2018 to 2024 the model is trained only on earlier years (for 2018: 2015 to 2017) and forecasts the return of every firm for the next year. A real investor never has later data, so neither do we.
- **Step 2, portfolios and costs.** Each July: take the firms of the pool, pick the 10% with the highest forecast, hold them equally weighted until the next June (firms that leave contribute until their last month), and charge trading costs on the share of the portfolio that changed. The floor versions first remove firms with a low Environmental score. The benchmark holds all firms equally.
- **Step 3, bootstrap.** Resample the 84 months with replacement 5,000 times and recompute the Sharpe difference each time, to see how much it moves by chance. The interval is the middle 95% of these differences.
- **Step 4, figure** of the cumulative return series.
- **Step 5, numbers for results.json** (four rules and the bootstrap).

- **Step 6, checks on what drives the result.** Random selections of the same size; every firm above the floor without a forecast; each holding year separately and the backtest without 2022; a survivors-only benchmark; formation year 2025 included; the bootstrap with blocks of months.
- **Final cell.** Collects everything from tasks (a) to (d), checks that no number is missing or infinite, and writes results.json with the group name Group_8 and the three members.

6. Questions you may be asked

Question	Answer
Why does the strategy beat the benchmark if the forecast has no skill?	It does not beat it in any way that can be told from chance: the interval of the Sharpe difference contains zero. The 0.6-point gap comes with a very different portfolio (94 names against 945).
Why did the floor not cost return?	With 19 to 47 names the result is dominated by which names happen to be picked. We make no claim that ESG screening adds return. The honest cost of the floor is concentration and trading.
Why start in 2018, and why leave out 2025?	The first forecast needs several earlier years to train on (2015 to 2017). Formation 2025 has only 6 months of returns and is left out; including it lowers every rule and the benchmark by 1.6 to 2.3 points a year and keeps the ranking ((d) Step 6).
Is the benchmark fair?	It is equal weight on all firms of the formation panel including firms that left, with no cost, as in the starter. Using survivors only would raise it from 8.77% to 9.22% a year (+0.46 points over 2018 to 2024; (d) Step 6).
How realistic are the costs?	A flat 20 bp each way is a simplification. Real costs vary by stock size and market conditions; the memo should say so.
Why bootstrap months?	To see how much the Sharpe difference moves by chance with only 84 months. Resampling blocks of consecutive months, which keeps dependence between months, gives intervals that are no wider, so the conclusion does not depend on it.

7. What to be careful about

- Do not present the 80th-percentile floor (11.4%) as an ESG result: all firms above the floor earn about the benchmark, the gap is one year (2022) and 19 names, and the interval contains zero.
- Do not pick the linear variant because it looks best: it is a labelled robustness check.
- The tuning of gradient boosting was done once on the training years of task (b), a simplification stated in the notebook.
- Report the benchmark next to every metric. Say that the sample is 84 months in one mostly rising market.
- Numbers come from the run of 4 November.

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Plain-language explainer

Ideas behind the coaching questions

Item	Details
Institution	Bern University of Applied Sciences (BFH)
Course	EAIF: AI for Finance, Autumn Semester 2026
Group	Group_8 (Beyeler Fabian, Lussi Laurin, Griessen Jan)
Document type	Background reading in everyday language, corrected 3 November 2026 (28 firms)

Purpose. Explains the ideas behind the project (leakage, formation year, baselines, regression, cross-validation) with small examples. No finance or statistics background needed.

What this document is. Each section explains the idea behind one question on the question sheet, in everyday words, with a small example. You do not need any finance or statistics background. The numbers come from the course data, the starter notebook and our own audit in the solution notebook.

The big picture in five sentences

- The course gives us **simulated** data about 1,376 companies over 14 years (fiscal years 2012 to 2025). No real firm is behind it.
- Every company publishes a report once a year. The year it covers is its **fiscal year**. The data has one row per company per year, plus one row per company per month for its stock return.
- An **ESG score** (0 to 100) says how responsibly a company behaves. We look at the **Environmental** part.
- We pretend to be an investor who picks stocks on **1 July** each year, holds them for 12 months, and asks: could we have predicted which stocks do well?
- The assignment tests whether we handle the data honestly. The main danger is using information the investor could not have had at the time. This is called **leakage**.

The four tasks of the assignment

- **(a) Audit:** check the data for errors (the course deliberately inserted some) and write a table: the problem, how we found it, how we fixed it, and what it changed in one number.
- **(b) Models:** predict each firm's return over the next 12 months with a simple model (linear regression) and with tree models, and compare them with a naive guess.
- **(c) ESG premium:** ask whether firms with a better Environmental score earned higher returns afterwards, by region and period. A **premium** is extra return earned by one group of firms.
- **(d) Backtest:** replay a strategy on past data as if we had traded it: each July buy the top-ranked 10% of firms (a **portfolio** is the set of stocks held), hold for 12 months and pay trading costs. Compare with buying all firms equally.

A **feature** is an input column the model uses to predict (for example profit margin). The **target** is the number it predicts.

1 Units and rescaling

Belongs to: Q1

A column has a **unit**. Here, Revenue and Market_Cap are in **USD million**. If one value was typed in USD *thousand* instead, it looks 1,000 times bigger than it should be. It is like one runner's time being recorded in seconds while everyone else's is in minutes.

What the audit found: **28 firms** (302 firm-years) report revenue and market cap about 1,000 times too large. We found them by checking that carbon, water and energy use grow with revenue inside an industry: for these 28 firms they sit about 1,000 times below that relation, so only revenue and market cap are off. A simple revenue cut-off finds only 22 to 27 of them (firm 406 sits just below the usual line). Example: Company_ID 30 has revenue of 1,441,650 in 2012. As USD million that would be USD 1.4 trillion, bigger than any real company. Divided by 1,000 it is 1,441.65, which fits the typical firm (median revenue about 2,300 to 2,400). **Market cap** (the total value of a firm's shares) is scaled the same way.

Option on the sheet	What it means	Risk
Divide by 1,000, keep the firms	We treat the values as a unit mix-up and repair them.	If they really are huge firms, we shrink them wrongly.
Keep as is	We treat them as real mega firms.	Size features (such as the logarithm of revenue, a transformation that shrinks big numbers) get extreme outliers: values far from all the others that pull averages.
Exclude the firms	We remove them completely.	We lose 28 firms (and the audit table should say so).

Why it matters: the audit table (task a) is graded against a list of planted problems. The course deliberately inserted errors into the data. A unit problem is a classic one, and the grader wants to see what we did and what it changed. The lecturer agreed that dividing by 1,000 is acceptable.

2 Near-duplicate rows and why they matter

Belongs to: Q2

A **duplicate** is a row that appears twice. Here the two rows are not exactly equal: Company 1, March 2020 has -27.7587 and -27.7549 (percent). That looks like one measurement stored twice with a tiny rounding difference.

Why it matters: a 12-month return is built by **compounding** the monthly returns, which means multiplying them. Each month is a factor: a return of -27.76% is the factor 0.7224. Take a firm that earned 0% in every other month. Its true yearly return is -27.76%. If the March row is counted twice, the factor is 0.7224 x 0.7224 = 0.5219, which gives **-47.8%**. One extra row makes the loss about 1.7 times larger (-48% instead of -28%).

The two values differ by at most 0.01 percentage points, so keeping the first row or averaging them gives practically the same result. Dropping both deletes the month entirely, which leaves a hole in that firm's year. In the starter, that firm-year would then have only 11 holding months and drop out of the model data (it needs 12), so it is the least attractive.

3 Total return, excess return and what a "target" is

Belongs to: Q3

The **target** is the number the model tries to predict. Here it is a firm's return over the 12 months after the portfolio is formed.

	Definition	Example
Total return	What an investor in the share earned over the period.	Firm earned +8% over the year.
Risk-free rate	What a safe account pays. In the data it is 0.1% to 0.4% per month, which is roughly 1.2% to 5% over a year.	Safe account paid about 1.2% (in a year at 0.1% a month).

	Definition	Example
Excess return	Total return minus the risk-free rate. It measures the reward for taking risk.	8% - 1.2% = about 6.8%.

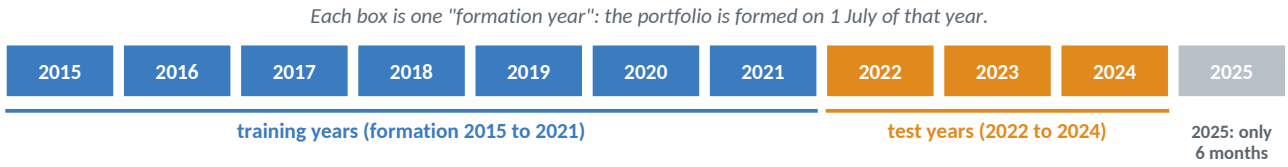
The starter notebook predicts the **total** return in task (b), and so does our notebook. Task (c) asks about the **excess** return. They are close in the training years (about 1 point apart on average) but about 5 points apart in the test years, because the risk-free rate rose after 2021. Every table in the memo should say which one it uses. The handout's decision-time rule names the total return as the target.

The baseline. The simplest possible forecast is "everyone earns the average of the training years". Its test error (RMSE, the typical size of a forecast error in percentage points) is 36.93 in the starter and 37.20 in our cleaned data. A model is only useful if it beats this number. In the starter, the random forest gets 37.31, so it does *not* beat the baseline, and the same is true for all our models. The assignment says that is a legitimate result, as long as we say so. Individual stocks swing a lot, so even the best forecasts are far off. What matters is the difference to the baseline.

4 Formation year, holding period and training years

Belongs to: Q4

The **formation year** is the year in which the portfolio is formed (on 1 July). The **holding period** runs from that July to the next 30 June. The starter fits the model on formation years 2015 to 2021 (training) and tests it on 2022 to 2024 (test). A random split would be wrong: a model could then learn from 2023 returns to "predict" 2021, which no real investor can do.



Why 2025 is special. Returns stop in December 2025. A portfolio formed in July 2025 can only be followed for 6 months, so its 12-month return does not exist yet. The starter's model leaves 2025 out (it needs 12 months). Its **backtest** keeps it (6 months), which is why the starter's backtest has 126 months. Our own backtest (task d) uses formation years 2018 to 2024 (84 months) and leaves 2025 out.

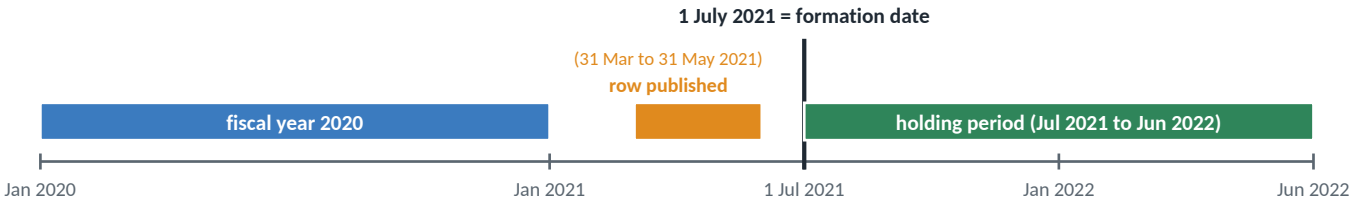
Walk-forward training (task d). In the backtest, the forecast for each formation year may only use *earlier* formation years. For 2018 the model trains on 2015 to 2017. For 2019 it trains on 2015 to 2018, and so on. That mimics a real investor, who never has future data. The first years have very little training data, so we start the backtest in 2018.

5 Fiscal year, formation year and calendar year

Belongs to: Q5

Three different "years" appear in the data, and mixing them up causes leakage.

- **Fiscal year t:** the year the company reports on. Example: fiscal year 2020.
- **Formation year:** the year in which we may use that report. A fiscal-year-2020 row is published between 31 March and 31 May 2021, so it is used for the portfolio formed on 1 July 2021. Formation year = fiscal year + 1.
- **Calendar year of the return:** the months in which the stock moved. The holding period July 2021 to June 2022 straddles two calendar years, so this option is ambiguous.



The fiscal-year-2020 numbers may be used on 1 July 2021. Nothing published later may be used.

Why the question matters. Task (c) compares "before 2021" and "from 2021". The score for fiscal year 2020 belongs to the formation year 2021. If the split is defined on the *fiscal* year, that score falls in the "before" group. If it is defined on the *formation* year (our choice, confirmed in the coaching), it falls in the "from 2021" group. The assignment says the data contains one genuine relationship, which should show up in one cell of the region-by-period table, so moving a year of data between the two periods can hide or dilute it.

6 Standardising the score (and the task (c) method)

Belongs to: Q6

Standardising (z-score). Subtract the average and divide by the spread: z = (score - average) / standard deviation. After this, the average is 0 and one unit means "one standard deviation higher". In the data the within-year spread of the Environmental score is about 17 points, so z = +1 means about 17 points above the year's average. We standardise within each year so that a year with generally higher or lower scores does not distort the comparison.

The option choice. "Within year, across all firms" compares a firm with everybody. "Within year and region" compares it only with its regional peers. Because task (c) reports results by region, standardising within region would change what "one unit" means in each region. Example: if the spread in Europe is 10 points and in Asia 25, one unit would mean 10 points in Europe but 25 points in Asia, so the units would differ between regions.

Only needed later, not for the question: the method behind task (c)

Regression with industry-by-year controls. A regression fits a line through a cloud of points. Its slope (the **coefficient**) says how much the return changes when the score goes up by one unit, holding other things fixed. "Industry-by-year controls" means we compare each firm only to firms in the same industry in the same year (for example Energy firms in 2019). This stops us from mistaking an industry boom for an ESG effect.

Standard errors. A coefficient is only an estimate. The standard error says how uncertain it is. If the estimate is less than about two standard errors from zero, we cannot tell it apart from noise. Two ways to get honest standard errors: **clustering by company** (we tell the software that the rows of one company belong together, because one company's years resemble each other and do not count as separate evidence; this is unrelated to clustering in machine learning), or the **bootstrap** (draw rows at random from our own data, with replacement, thousands of times, and see how much the answer jumps).

7 Leakage and the Analyst_Outlook column

Belongs to: Q7

Leakage means the model sees information it would not have in real life. It is like a student who studies with the answer sheet: great marks, no real skill. The decision-time rule is the test: *was this value in a file that existed on 1 July, with the value it had then?*

Our finding. Analyst_Outlook is a score from 1 to 5. We measured its rank correlation with the return in several 12-month windows. A **rank correlation** (Spearman) asks whether firms with a higher rank on one variable also tend to have a higher rank on the other. It runs from -1 to +1, and 0 means no link.

Here t is the fiscal year of the row. Example for the portfolio formed on 1 July 2021 (the fiscal year 2020 row):

12-month window compared with Analyst_Outlook	Rank correlation
The 12 months after formation, Jul 2021 to Jun 2022 (the thing we try to predict)	**0.68**
The 12 months before formation, Jul 2020 to Jun 2021	-0.01
The 12 months before that, Jul 2019 to Jun 2020	0.01

Real analyst ratings predict returns only weakly. A column that agrees so strongly with the future and with no other window behaves as if it was stamped with future information. A model that used it would look great on paper and be useless in practice: in our data a random forest reaches a test error of 29.1 with it against 37.6 without it (baseline 37.2).

Option on the sheet	What it means
Yes, that is enough	We report it in the audit table (how found, that we excluded it) and list it as "excluded" in the feature table. The feature table shows every candidate feature and the date from which it is known.
Also show a model with it included	A warning demonstration: it shows how misleadingly good a leaking model looks. The options add to each other: tick all that apply.

8 The "effect on one headline number"

Belongs to: Q8

A **headline number** is one figure that summarises a result, for example the average 12-month return, the model's test error (RMSE), or the Sharpe ratio. For each problem in the audit table, we show what the number was *before* and *after* the fix. Example row: "Near-duplicate returns; found by counting repeated firm-months; fixed by keeping one row; firm-years with a full 12 holding months went from 9,002 to 9,182."

The **Sharpe ratio** is the average excess return divided by its ups and downs (volatility), scaled to a year. It measures return per unit of risk. Higher is better.

9 Page and slide limits

Belongs to: Q9

This is a rule question, no concept needed. The handout says "3-page memo (plus appendix of any length)" and "at most 8 slides, 10-minute talk". The lecturer confirmed that tables and figures count towards the 3 pages and that the number of slides and the software are free; the talk is 8 minutes plus about 2 minutes of questions.

10 Libraries, Colab and run time

Belongs to: Q10

Colab is Google's free online notebook: our code runs on Google's computers. A **library** is a toolbox of ready-made code; **pip** is the command that installs one. **scikit-learn** has the models of task (b). **statsmodels** fits regressions and prints standard errors, which task (c) needs. Our notebook runs top to bottom in a few minutes on Colab (the model tuning is the slowest step).

Optional background for tasks (b) to (d)

Name	In plain words
Linear regression	Finds one weight per input and adds them up (like a recipe). Easy to read, can only draw straight-line relationships. The handout asks for standardised inputs, which makes the weights comparable.
Random forest	Builds hundreds of decision trees (flowcharts of yes/no questions such as "is profit margin above 10%?") and averages their answers.

Name	In plain words
Gradient boosting	Also trees, but built one after another. Each new tree tries to fix the mistakes of the previous ones.
Overfitting	The model memorises the training data (including its noise) and does badly on new data.

Tuning and cross-validation. A model has **settings** chosen by us, such as how deep the trees may grow (hyperparameters). We choose them with **cross-validation**: cut the training data into 5 parts, train on 4, check on the fifth, and rotate. Our folds keep all rows of one company in the same part, so the model cannot recognise a company it saw in training. Inside the training years the parts mix different years, so the cross-validation score is a little optimistic. That is why the test years come last and are used once.

Our result (cleaned data)	RMSE (percentage points)
Naive baseline on the test years	37.20
Linear regression, test	38.10
Random forest (tuned), test	37.93
Gradient boosting (tuned), test	37.84
Cross-validation (training years), linear regression	28.88

Cross-validation looks much better (28.9) than the test (37 to 38). The starter's own text blames a market downturn in the test years, but the data shows the opposite: the average 12-month return was **+21.6%** in the test formation years 2022 to 2024 against **+0.9%** in the training years, and only 2018 (+24%) was as high in training. The training average is therefore far too low for the test years.

Tune once vs re-tune each year (task d). The handout accepts both. We tune gradient boosting once on the training years of task (b), which is a simplification because the setting uses years the early forecasts should not have seen. Linear regression has nothing to tune.

11 Use of AI tools

Belongs to: Q11

The lecturer's answer in the coaching: AI tools are encouraged. The handout also states that every group member must be able to explain every cell of the notebook, so it is best to keep a short log of where AI helped.

12 Notebook version

Belongs to: Q12

The notebook deadline will be communicated by the lecturer and will be close to the hand-in date in November. The handout gives the dates: 27 November for the memo, notebook and peer evaluation; 7 December for slides; 8 December for presentations.

A Quick glossary

Terms used in several sections

Term	Meaning
Fiscal year	The year a company's yearly report covers.
Leakage	A model uses information that would not have been available at decision time.
Backtest	Replaying a strategy on past data as if it had been traded.
Baseline	A deliberately simple forecast (here: the training average) that every model must beat.
Compounding	Multiplying monthly return factors to get a yearly return.

Term	Meaning
Volatility	How much returns go up and down; technically the standard deviation of returns, usually per year.
Point-in-time	Using only what was known on the decision date, with the value it had then.
Formation date	1 July each year: portfolios are picked and held until the next 30 June.
Survivorship bias	Testing only on firms that still exist, which hides the ones that failed. The file universe_current.csv has only survivors (987 of 1,376 firms).
Restated scores	ESG scores as a download today shows them, revised with hindsight, so they are not what an investor knew back then.
RMSE	Typical size of a forecast error, in the units of the target (percentage points of return). Lower is better.
R squared	Share of the variation the model explains. 0 is as good as the average; negative is worse than the average.
Bootstrap	Draw rows at random from our own data, with replacement, thousands of times, to see how much a result moves by chance.
Sharpe ratio	Average excess return divided by volatility, scaled to a year: return per unit of risk.
Turnover	Share of the portfolio replaced at each rebalancing. Each unit costs 20 basis points (0.20%) in the starter's convention, charged on the sell and the buy.

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Group plan

Timeline, roles and internal targets

Item	Details
Institution	Bern University of Applied Sciences (BFH)
Course	EAIF: AI for Finance, Autumn Semester 2026
Group	Group_8 (Beyeler Fabian, Lussi Laurin, Griessen Jan)
Document type	Project plan written 3 November 2026
Key dates	Coaching Tue 24 Nov 2026; memo, notebook, results.json and peer evaluation Fri 27 Nov 2026, 23:59; slides Mon 7 Dec 2026, 23:59; presentation Tue 8 Dec 2026

> **Purpose.** The plan of work from the start of the project to the presentation on 8 December, with roles and internal targets.

1. Key dates

Date	What
Tue 3 Nov (today)	We are in week 1 of the plan
Tue 24 Nov	Coaching with the lecturer: review of the notebook state and the draft slides
Fri 27 Nov, 23:59	Hard deadline on Moodle: memo (PDF), notebook (.ipynb), results.json, peer evaluation. Internal target: submit **Thu 26 Nov evening**.
Mon 7 Dec, 23:59	Final slides (PDF), Moodle or email to the lecturers
Tue 8 Dec	Presentation: 8 minutes plus about 2 minutes of questions

The lecturer will still communicate the notebook deadline, which will be close to the hand-in date in November. If it is earlier than 27 Nov, the freeze date below moves with it.

2. Status on Tuesday 3 November

Done	Still open (week 0 items that slipped)
Handouts and data dictionary saved in the repo. Coaching question sheet and plain-language explainer written. Coaching answers recorded. Task (a) drafted in the solution notebook (audit, treatments, formation panel; reviewed by Claude, waiting for your review in Colab). Starter notebook runs.	Roles confirmed by all three. Group name chosen. Notebook copy created. Both teammates have read the explainer.

3. Roles (proposal until all three confirm)

Person	Handout roles	Does
You (notebook owner)	Data lead + modelling lead	Owns the notebook, which Claude drafts. Reviews and runs it in Colab. Writes results.json.
Teammate B	Writing lead	The memo in Word, including the section "What would change with real data". Fills in the peer evaluation.

Person	Handout roles	Does
Teammate C	Analysis lead + slides	Turns notebook results into tables and figures, and builds the PowerPoint (6 slides for 8 minutes).

The handout says every member must explain every cell, and that the peer evaluation can adjust individual scores. Neither teammate codes, so each task ends with a short walkthrough and a plain-language guide (see section 8).

4. Decisions from the coaching (27 Oct)

The lecturer said there is a lot of leeway: any method is fine if it is consistent and we can explain why.

- Revenue unit problem: divide revenue and market cap by 1,000. Near-duplicate returns: keep one row per company-month.
- Target: total 12-month return in (b), excess return in (c). Period split in (c) by formation year. Environmental score standardised within year across all firms.
- Formation year 2025 in the backtest: either choice is fine, we must be able to defend it. Decided (4 Nov): left out of our backtest, because it has only 6 months of returns; including it lowers every rule and the benchmark by 1.6 to 2.3 points a year and keeps the ranking (notebook (d) Step 6).
- Analyst_Outlook: excluded from all models, listed in the audit table and in the feature table as excluded.
- Audit table: free choice of the headline number per problem, stating which one.
- Memo: the 3 pages include tables and figures. Slides: number and software are free, but the talk is only 8 minutes.
- Libraries: scikit-learn, statsmodels, possibly lightgbm. **AI tools are encouraged.**
- The lecturer suggested one slide that explains the audit: the problems we found and how we treated them.

5. Timeline by week

Week	Notebook (you, drafted by Claude)	Memo (B)	Slides (C)
1: 3 to 8 Nov	Finish setup (roles, group name, notebook copy). **Task (a):** audit and formation panel with 3 or more new dated features.	Memo skeleton, data section with the simulated-data statement and licence line. Preliminary audit paragraph from the lead list.	Slide skeleton (6 slides). Preliminary audit slide.
2: 9 to 15 Nov	**Task (b):** linear model, random forest, gradient boosting, compared with the baseline.	Pipeline and model sections. Outline of "What would change with real data".	Model comparison slide.
3: 16 to 22 Nov	**Task (c)** where ESG pays, **task (d)** backtest with floors and bootstrap. Then results.json and a full run in Colab. **Freeze Sun 22 Nov.**	Results sections. Complete memo draft by Sun 22 Nov.	ESG and backtest slides. Complete draft deck by Sun 22 Nov.
4: 23 to 27 Nov	Fixes after coaching. Teammate runs the notebook top to bottom in Colab.	**Tue 24 Nov coaching.** Then finalise the memo (Wed to Thu). Peer evaluation.	Adjust the deck after coaching feedback.
Submit	**Thu 26 Nov evening** (deadline Fri 27 Nov 23:59)		
5: 28 Nov to 8 Dec			Final slides by Mon 7 Dec (internal: noon). Two timed rehearsals (Sat 5 Dec and Mon 7 Dec). **Present Tue 8 Dec.**

6. Notebook steps in order

- **(a) Audit.** Check the leads: wrong revenue unit, near-duplicate returns, Analyst_Outlook leakage, survivorship (universe file), restated scores, non-random missing values, clipped scores. Look for anything else. For each problem record: how found, how treated, what changed in one number.
- **(a) Formation panel.** One row per company and formation year. Every feature gets the date from which it is known (is it in a file that existed on 1 July, with the value it had then?). At least 3 new features beyond the starter's.
- **(b) Models.** Linear regression on standardised features, random forest and gradient boosting. Tune with grouped cross-validation. Show train, CV and test side by side next to the naive baseline. The test set is used once. Say plainly if the ensemble does not pay off.
- **(c) ESG premium.** Previous year's Environmental score against the next 12-month excess return, by region and period (before 2021, from 2021), with industry-by-year controls and honest standard errors. Run on point-in-time and restated scores and explain the difference.
- **(d) Backtest.** Walk-forward forecasts, top decile held July to June, 20 bp cost per unit of turnover, ESG floors at the 50th and 80th percentile, bootstrap of the Sharpe difference to the equal-weight benchmark, on the full panel including firms that left.
- **results.json.** Fill the template, call write_results as the last cell, run the whole notebook top to bottom.

7. Internal targets for the team

Date	Target	Owner
Sun 8 Nov	Notebook task (a) done and reviewed. Results sheet (a) sent. Memo skeleton and slide skeleton ready. Walkthrough 1 (30 min).	You, B, C
Wed 11 Nov	Audit paragraph and audit slide drafted from results sheet (a).	B, C
Sun 15 Nov	Notebook task (b) done. Results sheet (b). Pipeline and model text, model slide. "What would change with real data" outline. Walkthrough 2.	You, B, C
Sun 22 Nov	**Freeze.** Tasks (c) and (d) done, results.json written, Colab run clean. Complete memo draft and complete deck. Walkthrough 3.	You, B, C
Mon 23 Nov	Whole-team read-through of memo and deck. List of open questions for the coaching.	All
Tue 24 Nov	**Coaching** with the lecturer.	All
Wed 25 to Thu 26 Nov	Fix findings, finalise memo, teammate runs notebook top to bottom in Colab, each person fills the peer evaluation. **Submit Thu 26 Nov evening.**	All
Fri 27 Nov, 23:59	Hard deadline (fallback only).	
Sat 5 Dec / Mon 7 Dec	Timed rehearsals (8 minutes). Slides submitted by Mon 7 Dec noon (deadline 23:59).	All
Tue 8 Dec	Presentation. Everyone can answer questions on every slide.	All

8. What each task must pass before it counts as done

- **Leakage check:** every feature is dated, and nothing used was unknown on 1 July of the formation year.
- **Baseline next to every metric:** the naive mean forecast for models, the equal-weight benchmark for backtests.
- **Figures:** labelled axes with units. **Tables:** the same numbers as results.json.
- **Claims checked:** a sentence like "beyond X" or "controlling for Y" is only written after the comparison was run (Prof. Matter's feedback). Every number gets its meaning for an investor, and magnitude words match the numbers.
- **Honest uncertainty:** say plainly when a result is not distinguishable from zero or from the baseline.

- **Handover:** a one-page results sheet and a plain-language guide for B and C, and a 30-minute walkthrough.

9. Support from Claude

- Drafts the notebook cells in group_project_AS2026/notebook/Group_Work_AS_2026_solution.ipynb (a copy of the starter, which stays untouched), runs them locally on the repo data to check they work, and commits. Nothing is pushed without asking.
- You review each block, run it in Colab and give feedback. You own every cell, so Claude explains each block in plain terms.
- Writes the results sheet and the plain-language guide for each task, with numbers checked against the data.
- Prepares an outline for "What would change with real data" (15 points) so B can start early.

10. Hours, risks and what to cut

Your budget is under 5 hours a week. From today to 27 Nov that is about 17 hours. Review and running time per task: (a) 4 to 5 h, (b) 3 h, (c) 3 h, (d) 4 h, final run and results.json 2 h, coaching 1 h, memo review 2 h. This is tight, so:

- Teammates take over anything that needs no code: running "Run all" in Colab, checking numbers in the memo against the notebook, formatting tables and figures.
- If behind, cut in this order: fewer bootstrap repeats, a smaller tuning grid, tuning once instead of re-tuning each year (the handout accepts both).
- Weeks 3 and 4 are the heavy part. Do not postpone the audit: tasks (b) to (d) depend on it.
- One person owns the notebook, so there are no merge conflicts. Teammates edit only the memo and the slides.

11. Open items to confirm

- Group name (needed in all filenames and in results.json).
- Full names of the three members and the final roles.
- Whether the talk is exactly 8 minutes (handout says 10).
- The notebook deadline once the lecturer communicates it.